上海科技大学
ShanghaiTech University

CS177 Bioinformatics: Software Development and Applications

Lecture 14: AlphaFold

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Spring, 2025

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Learning objectives

- After this lecture, you should be able to
  - Describe the historical and scientific background of the emergence of AlphaFold
  - Explain how AlphaFold 2 (AF2) works, e.g. the roles of multiple sequence alignment (MSA) and Transformer
  - Explain the main ideas of AlphaFold 3 (AF3)
  - Compare AF2 with AF3 to find some key differences
  - Describe at high level how diffusion models contribute to AF3
  - Point out some limitations of AF2 and AF3

Overview of protein structure

- The 3D structure of a protein determines its capacity to function
- In 1950s, Christian B. Anfinsen and others denatured ribonuclease, observed rapid refolding, and demonstrated that the primary amino acid sequence determines its three-dimensional structure
- We can study protein structures to understand problems such as:
  - the consequence of disease-causing mutations
  - the properties of ligand-binding sites, and
  - the functions of genes, etc.

Christian B. Anfinsen (1916-1995) American biochemist, Nobel Prize in Chemistry (1972)

Protein primary and secondary structures

- Primary structure

MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPKVKAHGKKVLGAFSD
GLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH

- Secondary structure

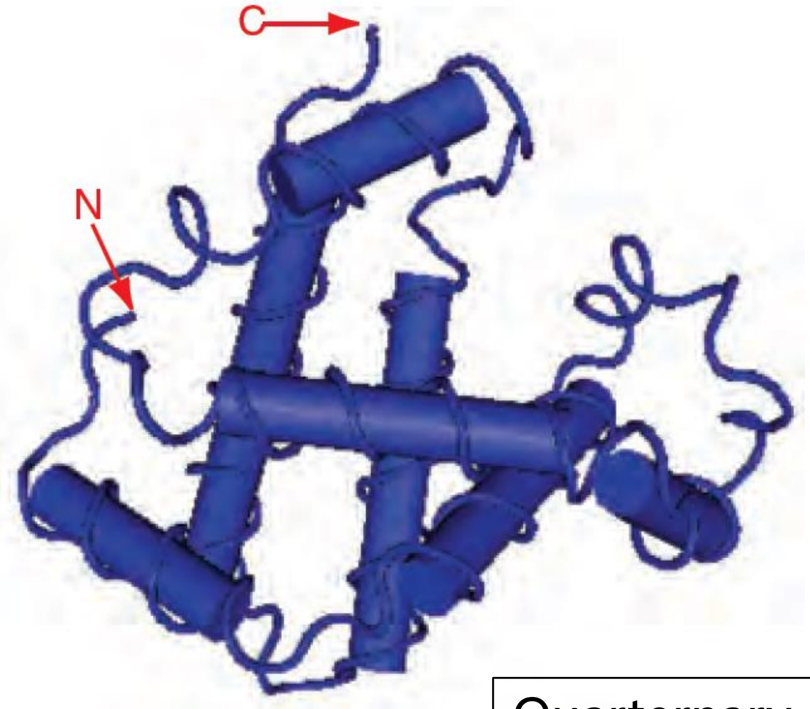
```
                     10        20        30        40        50        60        70
                      |         |         |         |         |         |         |
UNK_257900  MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPKVKAHGKKVLG
DSC         cccchhhhhhhhhhhhcccccchhhhhhhhhhhccccchhhhhhhhccccccccccccchhhhhhhhhhh
MLRC        cccccchhhhhhhhhhccccccccchhhhhhheeecccchhhhhccccccccccccccccccccchhhhh
PHD         ccccchhhhhhhhhhhcccchhhcchhhhhhheeeccchhhhhhhhcccchhhhheccchhhhhhhhhhhh
Sec.Cons.   ccccchhhhhhhhhhhcccccchcchhhhhhheeeccchhhhhhhhccccccccccccchhhhhhhhhhh

                     80        90       100       110       120       130       140
                      |         |         |         |         |         |         |
UNK_257900  AFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVAN
DSC         hhhhhhhhhhhhhhhhhhhhhhhhhhccccchhhhhhhhhhhhhhhhhccccccchhhhhhhhhhhhhhh
MLRC        hhhhhhhhhhhhhhhhhhhhhhhcccccccccchhhhhhhhhhhhhhhcccccchhhhhhhhhhhhhhhh
PHD         hhhhhhhhhhhhhhhhhhhhhhhhhccccchhhhhhhhhhhhhhhhhhhccccchhhhhhhhhhhhhhhh
Sec.Cons.   hhhhhhhhhhhhhhhhhhhhhhhhhcccccchhhhhhhhhhhhhhhhhcccccchhhhhhhhhhhhhhhh

UNK_257900  ALAHKYH
DSC         hhhhccc
MLRC        hhhhccc
PHD         hhhhhcc
Sec.Cons.   hhhhccc
```

Results from three secondary structure programs are shown, with their consensus:

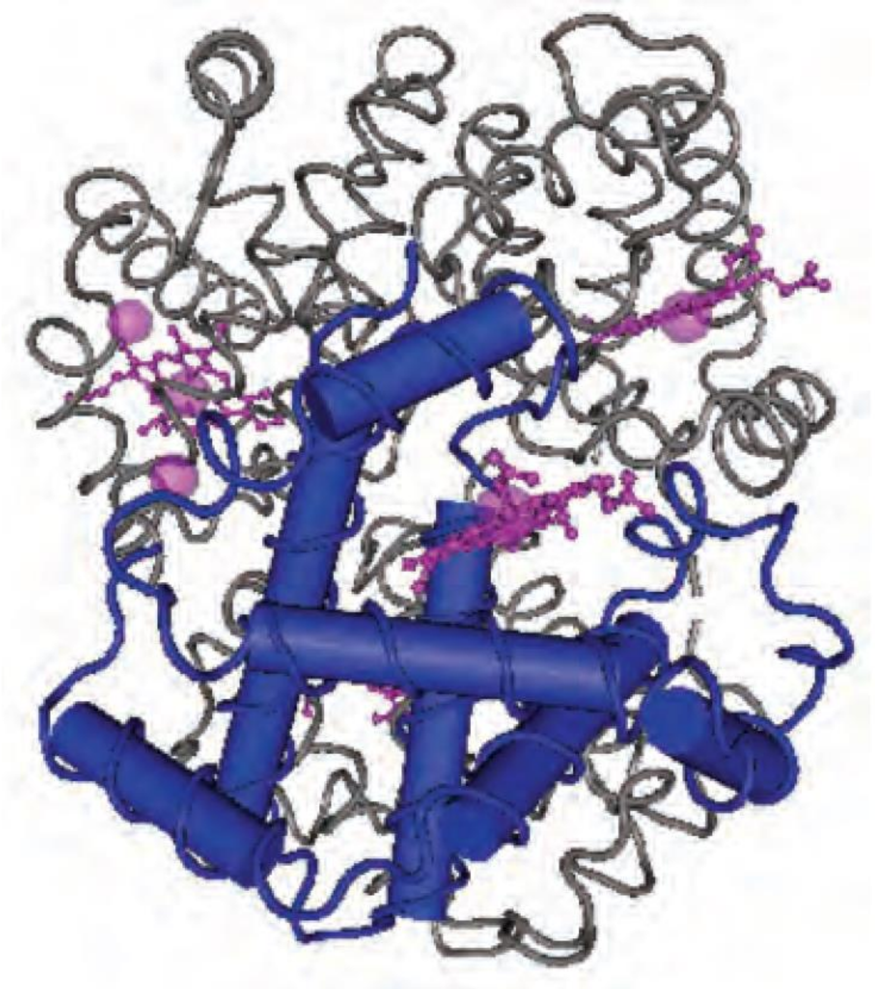
- **h**: alpha helix
- **c**: random coil
- **e**: extended strand

Protein tertiary and quaternary structures

Tertiary structure



Quarternary structure



Quarternary structure: the four subunits of hemoglobin are shown (with one beta globin chain highlighted) as well as four noncovalently attached heme groups.

Tertiary protein structure: protein folding

Experimental determination

- X-ray crystallography
- Nuclear magnetic resonance (NMR)
- Cryogenic electron microscopy (cryo-EM)

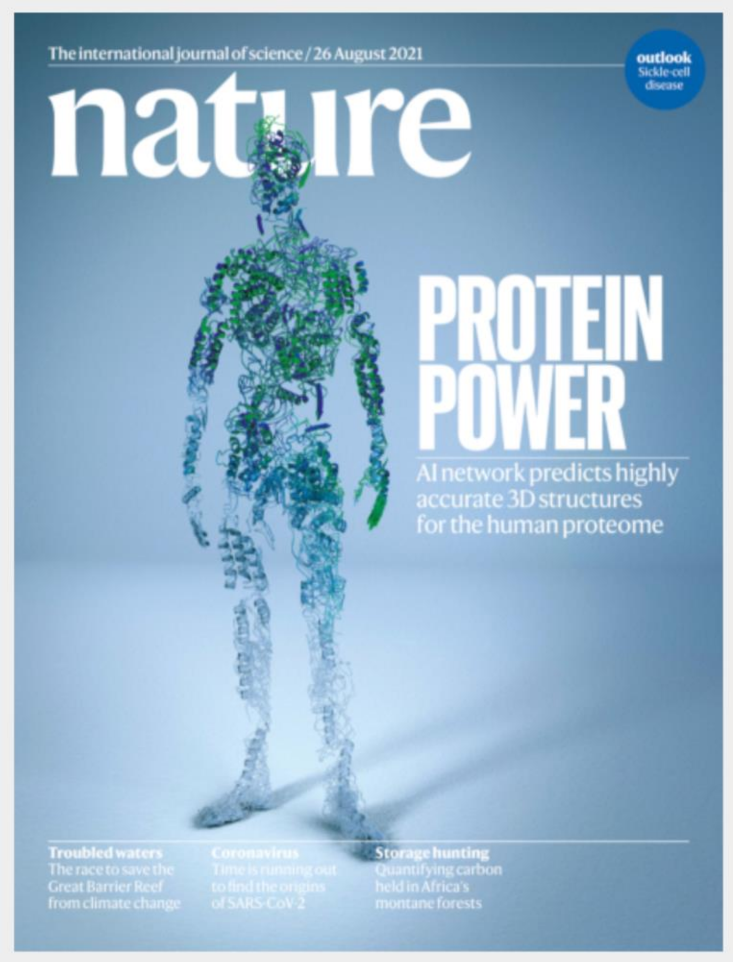
Computational prediction

- Homology modeling (comparative modeling)
- Fold recognition (threading)
- *Ab initio* prediction (template-free modeling)

Success of deep learning in biology

	Protein structure prediction	Protein function prediction	Genome engineering	Systems biology and data integration	Phylogenetic inference
Paradigm shifting	✓				
Major success		✓	✓		
Moderate success				✓	
Minor success					✓

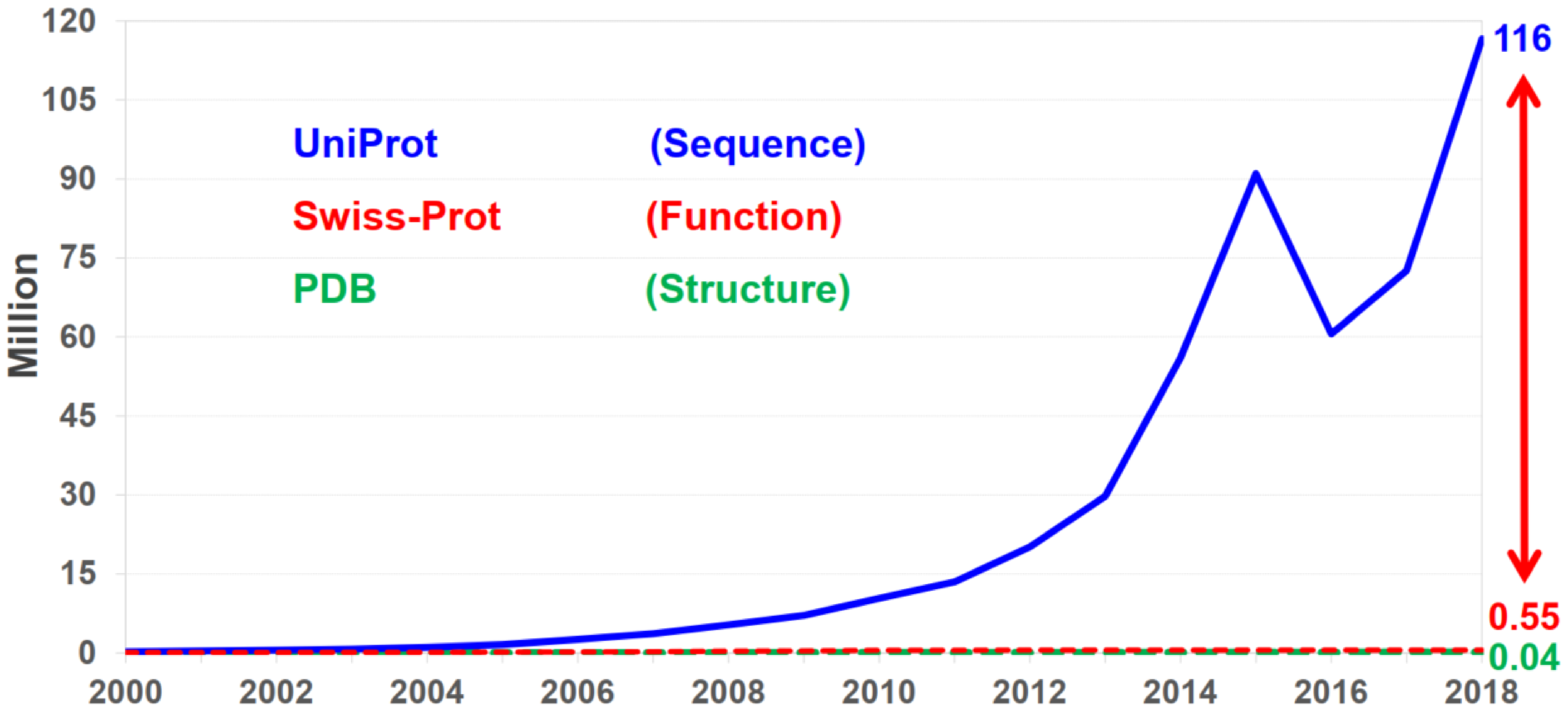
Sapoval et al. Nat. Commun. 2022



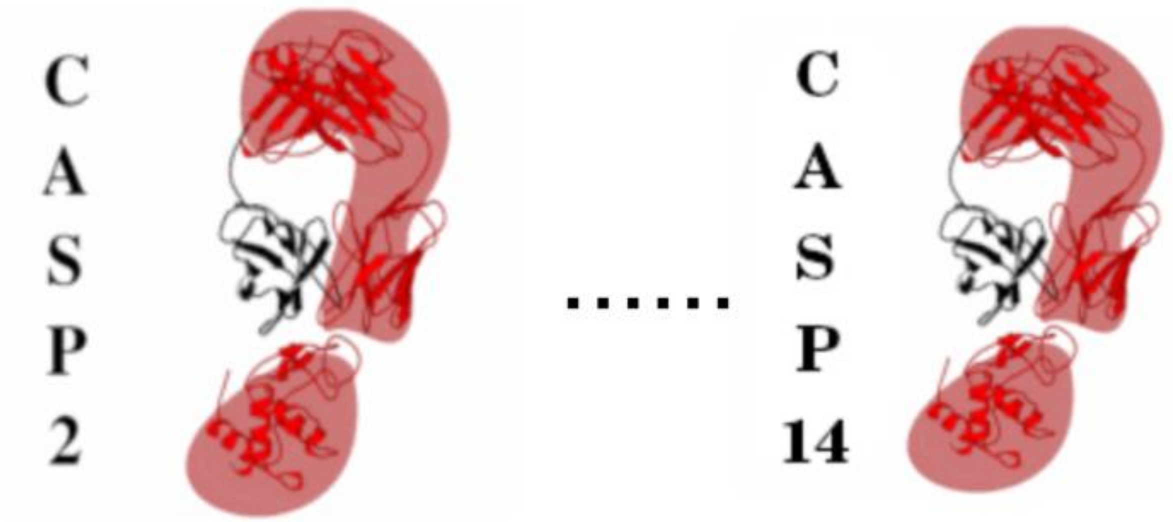
Cover image: DeepMind’s AlphaFold 2

AlphaFold 2

Sequence-structure/function gap keeps increasing: need computational prediction



The Critical Assessment of protein Structure Prediction (CASP)



1994 1996 2020

CASP1	1994
CASP2	1996
CASP3	1998
CASP4	2000
CASP5	2002
CASP6	2004
CASP7	2006
CASP8	2008
CASP9	2010
CASP10	2012
CASP11	2014
CASP12	**2016**
CASP13	2018
CASP14	2020

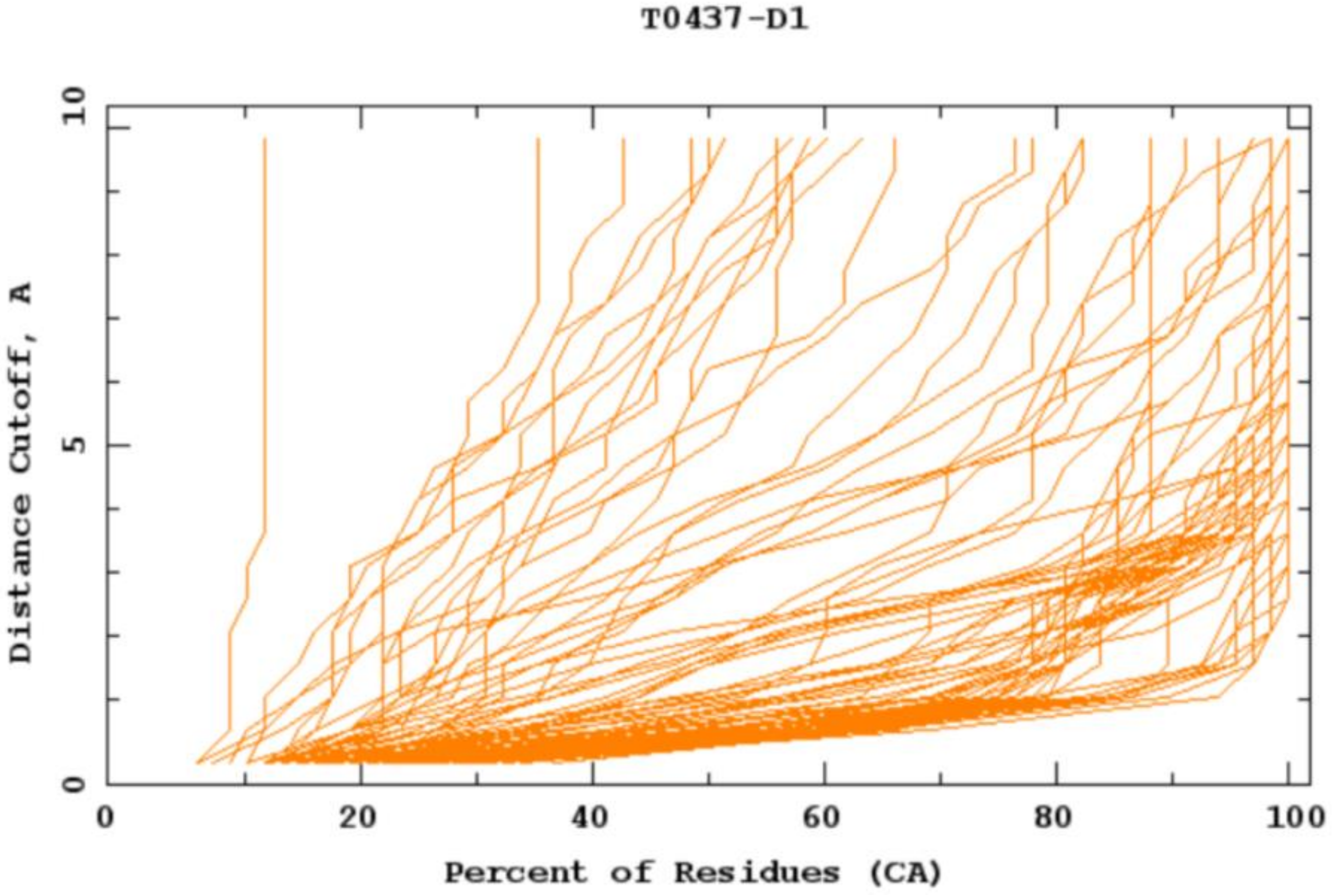
http://www.predictioncenter.org/

Evaluation of prediction by GDT_TS

GDT_TS = (GDT_P1 + GDT_P2 + GDT_P4 + GDT_P8)/4

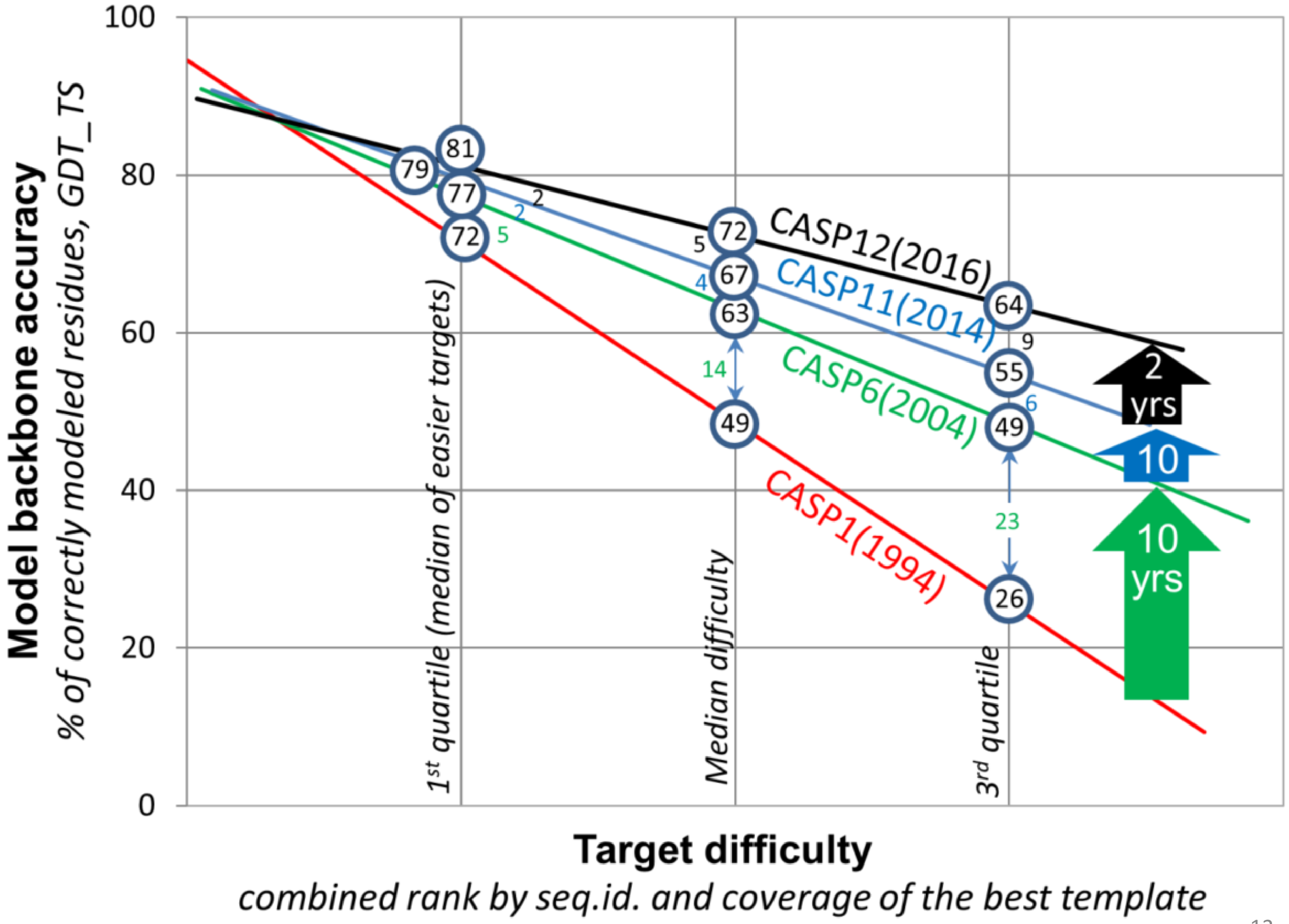
GDT_TS: GlobalDistanceTest_TotalScore

where GDT_Pn denotes percent of residues under distance cutoff <= nÅ



蛋白结构预测曾经十年停滞

2016年又重新爆发



Revolution starts from CASP12 (2016)

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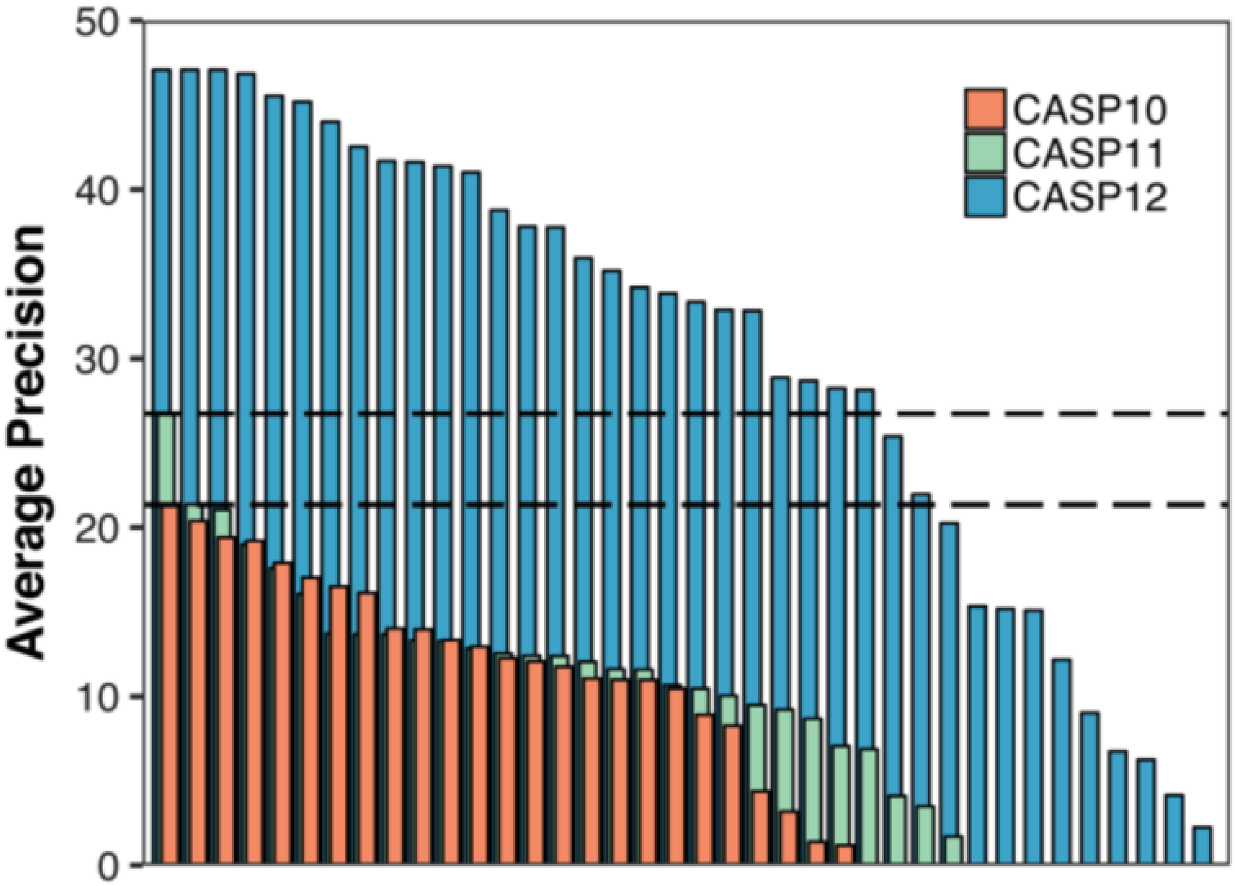
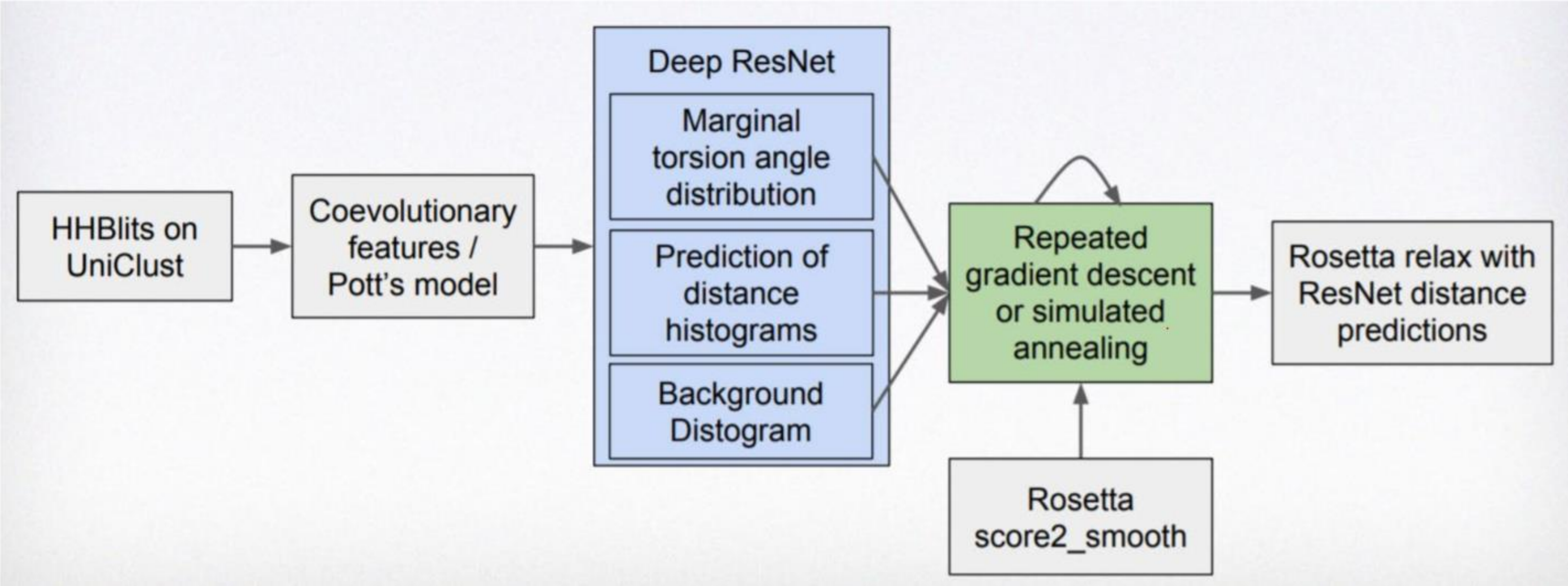
Contact map prediction represented by RaptorX-Contact



许锦波

清华大学卓越访问教授、美国芝加哥丰田计算技术研究所教授、分子之心创始人

No. 1 of CASP13: AlphaFold (version 1)



AlphaFold2 高精度破译人类蛋白质结构

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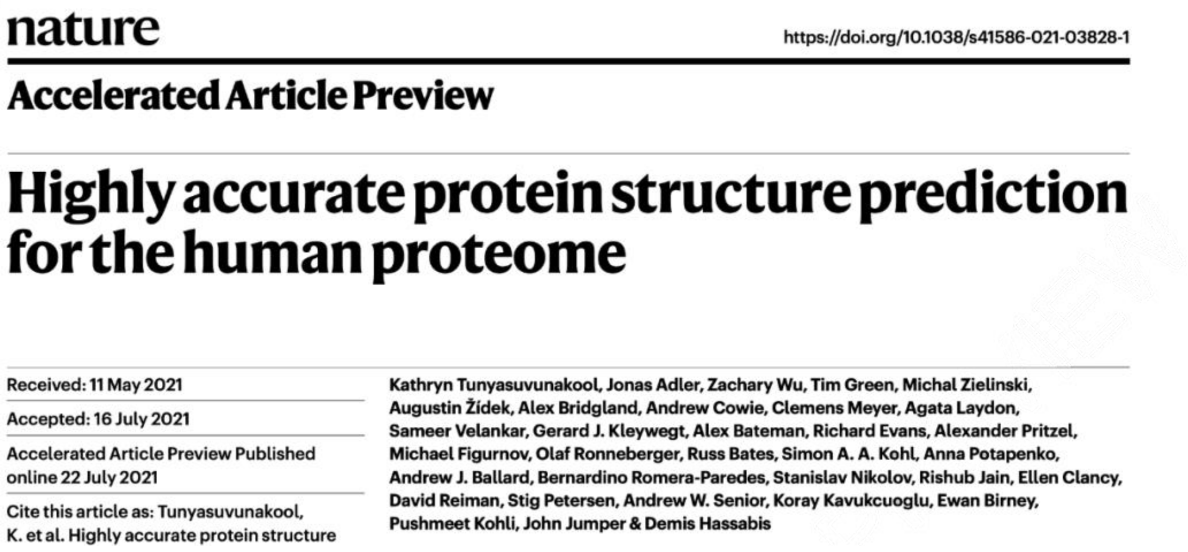

nature

https://doi.org/10.1038/s41586-021-03828-1

Accelerated Article Preview

Highly accurate protein structure prediction for the human proteome

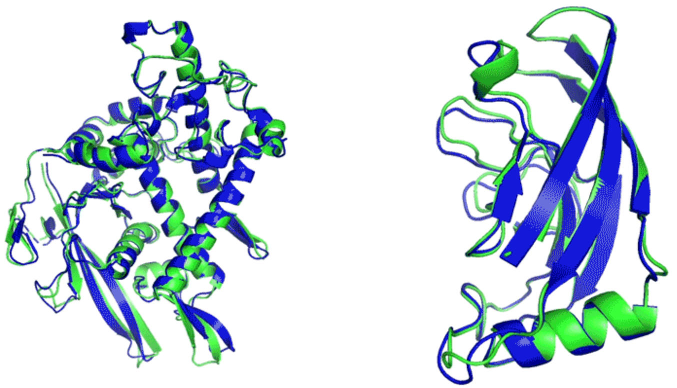
Received: 11 May 2021

Accepted: 16 July 2021

Accelerated Article Preview Published online 22 July 2021

Cite this article as: Tunyasuvunakool, K. et al. Highly accurate protein structure

Kathryn Tunyasuvunakool, Jonas Adler, Zachary Wu, Tim Green, Michal Zielinski, Augustin Žídek, Alex Bridgland, Andrew Cowie, Clemens Meyer, Agata Laydon, Sameer Velankar, Gerard J. Kleywegt, Alex Bateman, Richard Evans, Alexander Pritzel, Michael Figurnov, Olaf Ronneberger, Russ Bates, Simon A. A. Kohl, Anna Potapenko, Andrew J. Ballard, Bernardino Romera-Paredes, Stanislav Nikolov, Rishub Jain, Ellen Clancy, David Reiman, Stig Petersen, Andrew W. Senior, Koray Kavukcuoglu, Ewan Birney, Pushmeet Kohli, John Jumper & Demis Hassabis



T1037 / 6vr4
90.7 GDT
(RNA polymerase domain)

T1049 / 6y4f
93.3 GDT
(adhesin tip)

● Experimental result
● Computational prediction

2020年，DeepMind的AlphaFold2 (AF2) 在CASP14中取得了碾压性成绩：

- **大幅提高全原子蛋白结构预测的置信度和覆盖率**
- **蛋白结构的准确预测带来了高质量的生物学假设**
- **AI 驱动生物学研究的时代开始了**

CASP14 AF2预测样例展示：
绿色为真实结构，蓝色为预测结构。
GDT分数大于90（满分100）可以认为和真实结构差别不大。

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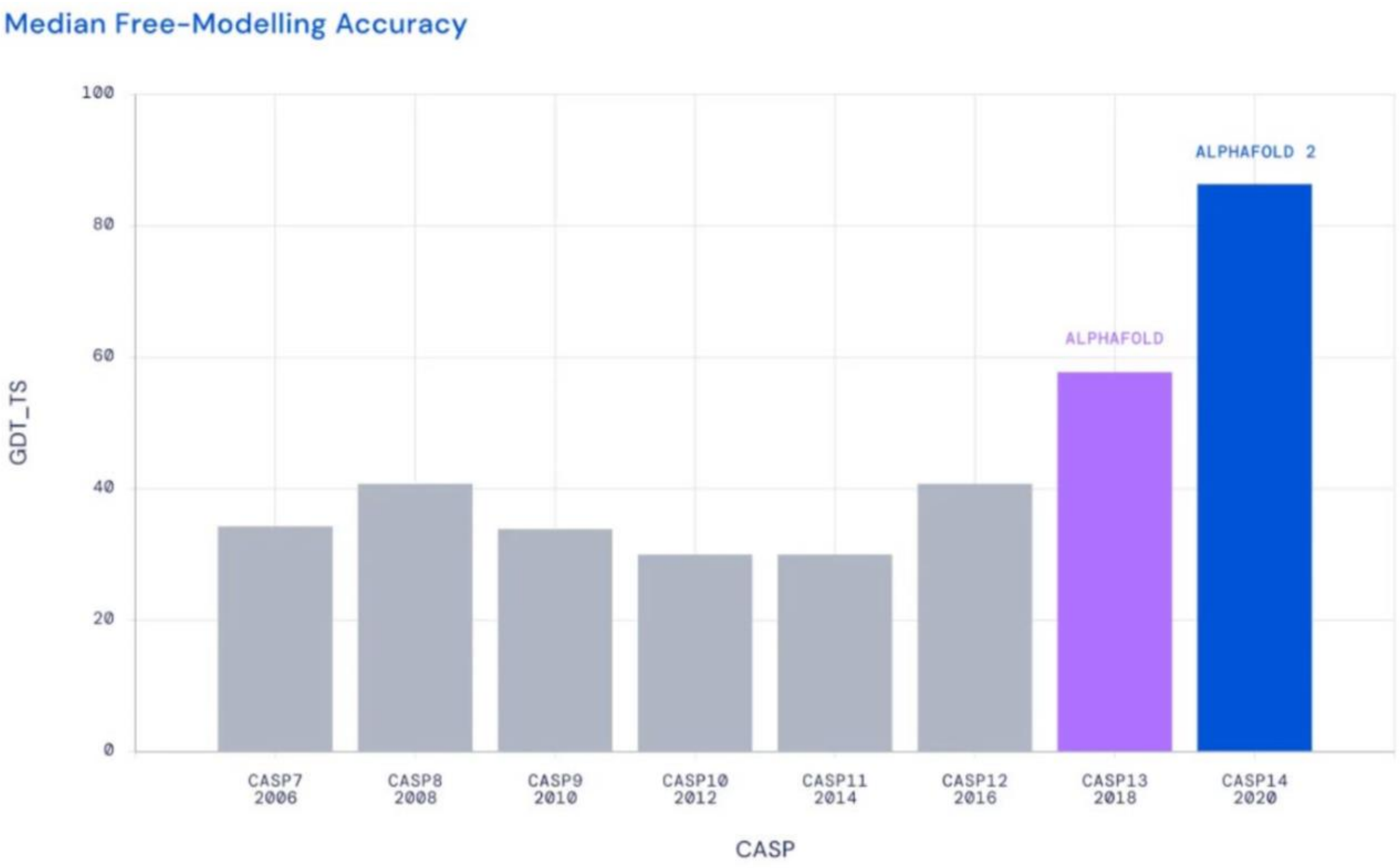
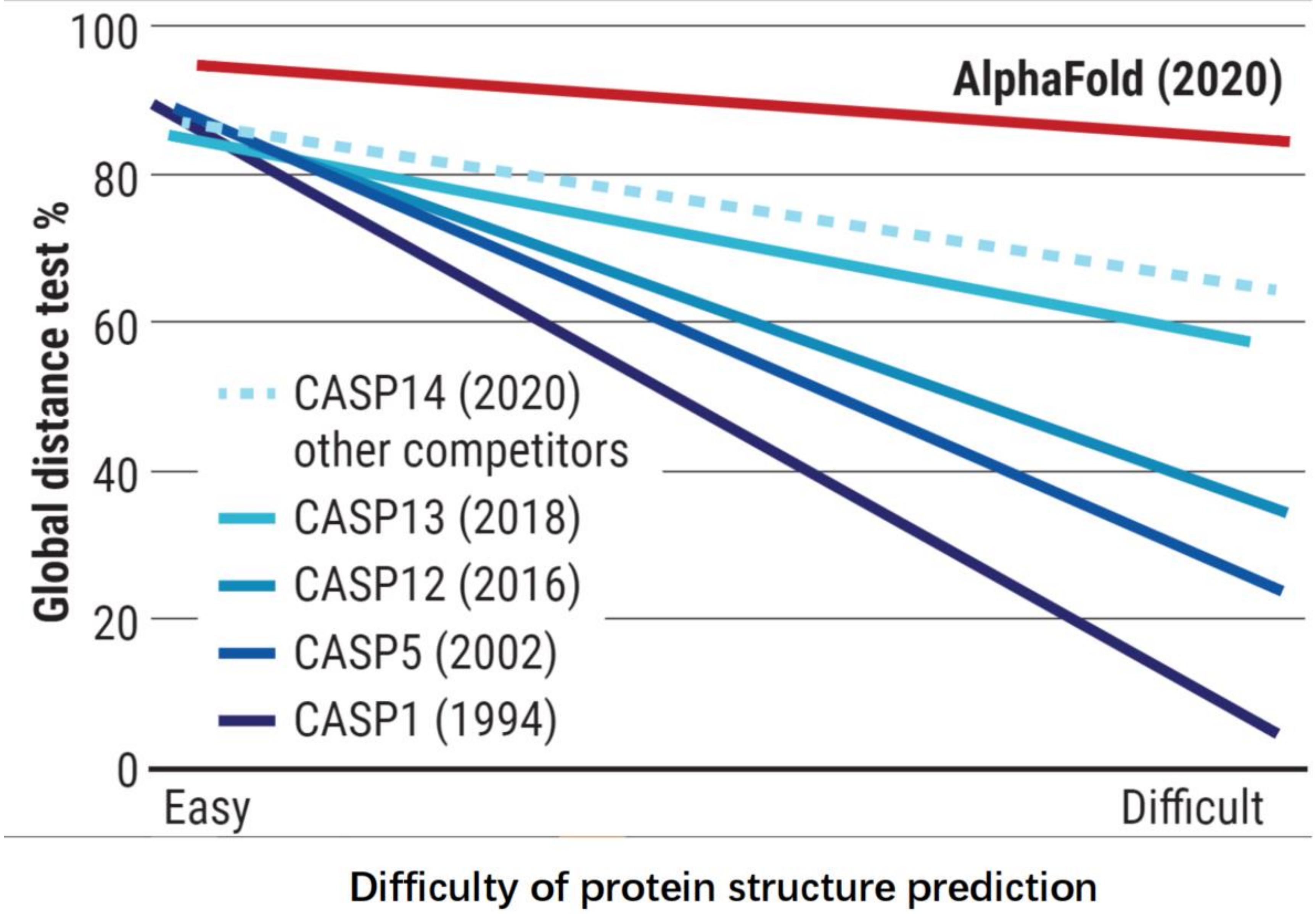
AlphaFold2’s accuracy is near experimental measurement



AF2 performs well even for difficult proteins



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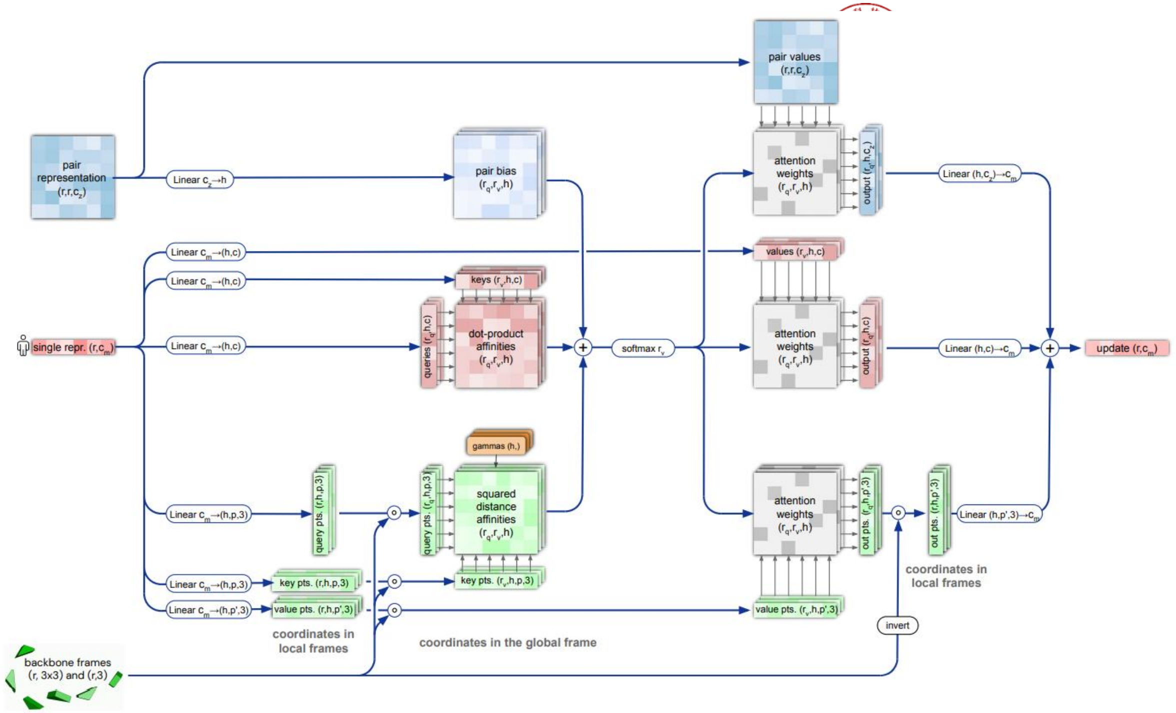
AlphaFold2 paper

- From Google DeepMind
- "Highly accurate protein structure prediction with AlphaFold" (Jumper et al. Nature 2021)
- Comprehensive description of AlphaFold2
  - 60 pages of supplementary materials
  - 32 algorithms
- Alongside it DeepMind open sourced the code:
  - Code: github.com/deepmind/alphafold
  - Colab: dpmd.ai/alphafold-colab



Algorithm 31 Generic recycling training procedure

def RecyclingTraining($\{\text{inputs}_c\}, N_{\text{cycle}}$) :

$N' = \text{uniform}(1, N_{\text{cycle}})$ *# shared value across the batch*

outputs $= \mathbf{0}$

 # Recycling iterations

for all $c \in [1, \dots, N']$ **do** *# shared weights*

    outputs $\leftarrow$ stopgrad(outputs) *# no gradients between iterations*

    outputs $\leftarrow$ Model(inputs_c, outputs)

end for

return loss(outputs) *# only the final iteration's outputs are used*

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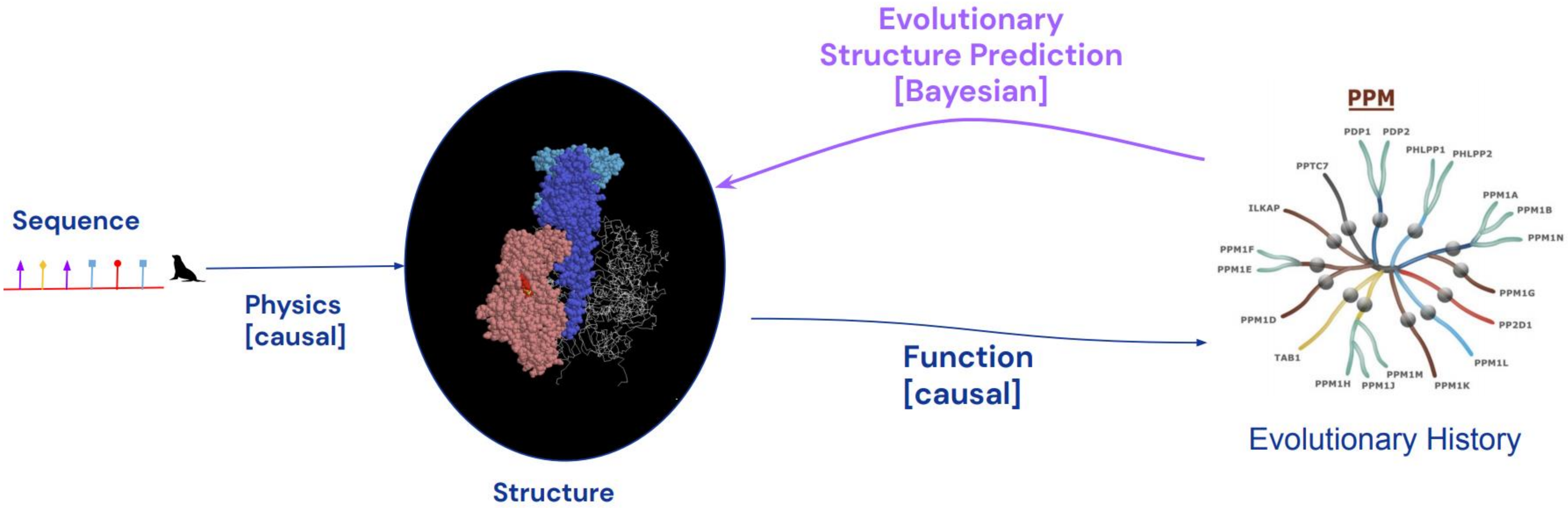
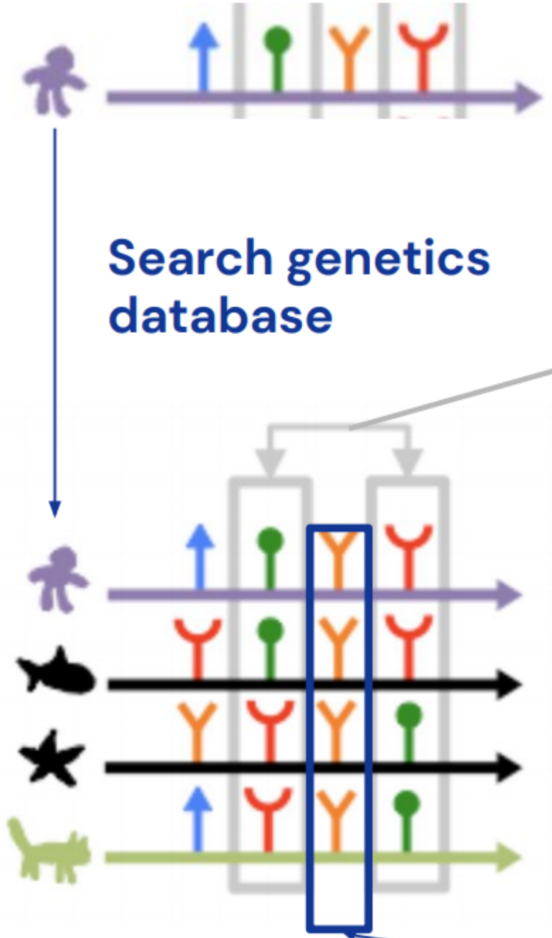
Determining structure from evolution – Intuition

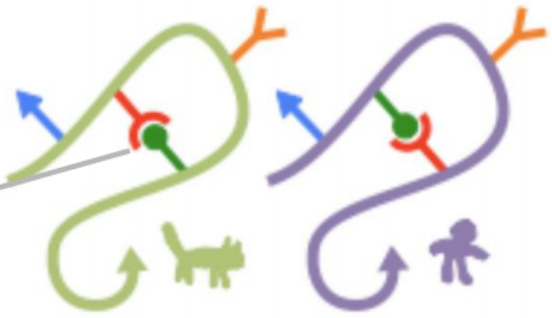


Source of slides: Google DeepMind

Determining structure from evolution – Intuition (continued)



Multiple Sequence Alignment (MSA)



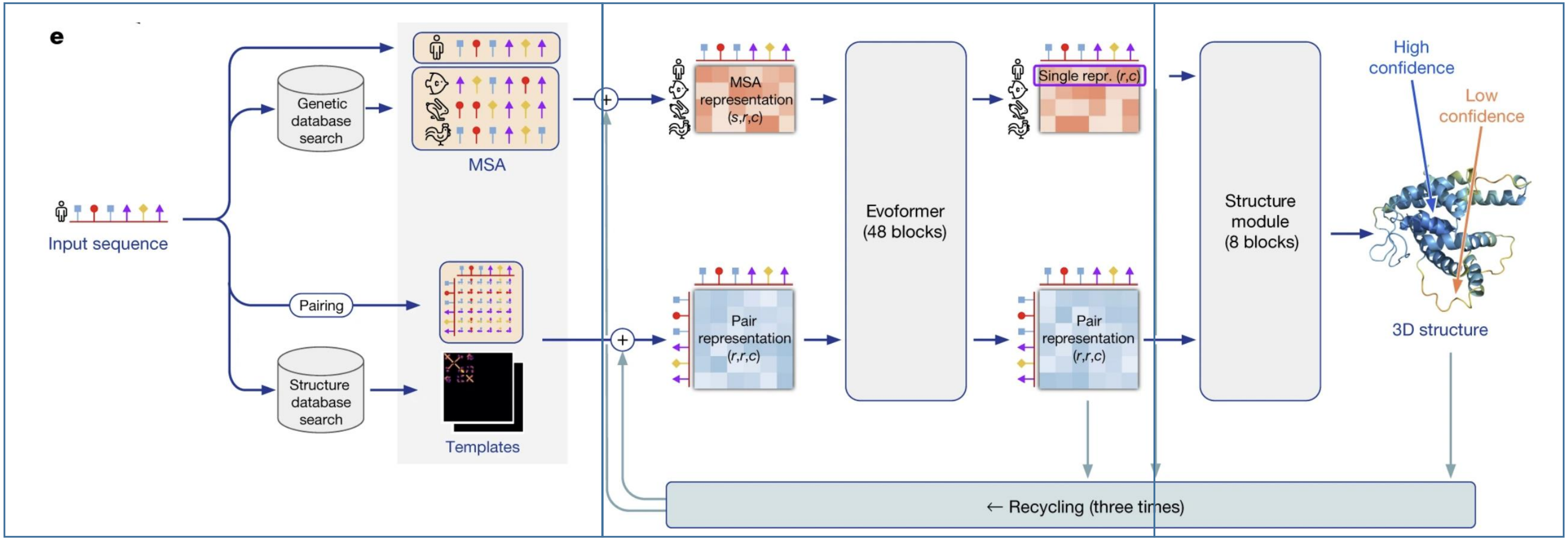
Co-evolution: residues in contact tend to mutate together.

(mutation of a single residue breaks the contact and the organism with the mutated protein does not survive)

Evolution conserves some properties like hydrophobic/hydrophilic amino acids on the "inside"/"outside"

Coevolution cartoon by Sergey Ovchinnikov (https://jgi.doe.gov/seeking-structure-metagenome-sequences/cartoon-coevolution-sergey-o/)

AlphaFold2的模型结构



AF2架构可以分为3个模块：

1. **Feature Embedder**：用于处理输入的蛋白序列，MSA和结构模板
2. **共进化蛋白语言模型**：称为Evoformer（Evolutionary Transformer）
3. **结构生成**：本质上是一个3D等变Transformer

Inputs

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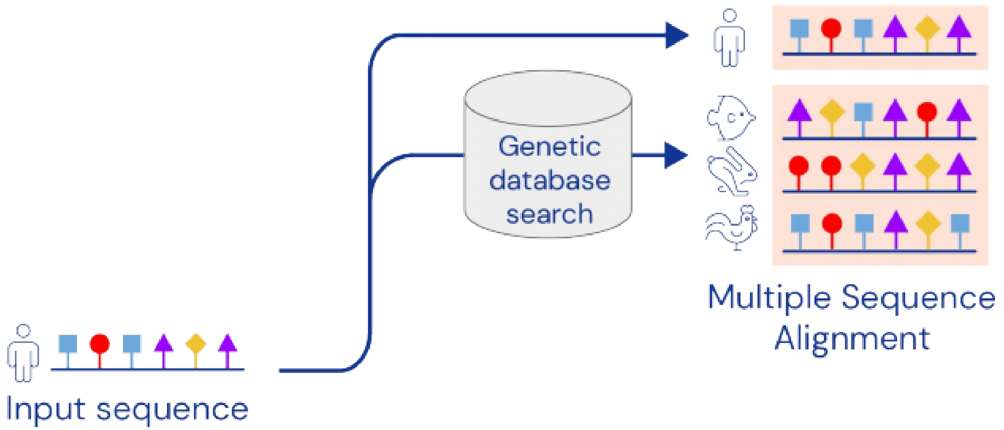

A key AlphaFold input is the MSA, containing sequences evolutionarily related to the target. Related sequences are found using standard tools and public databases.

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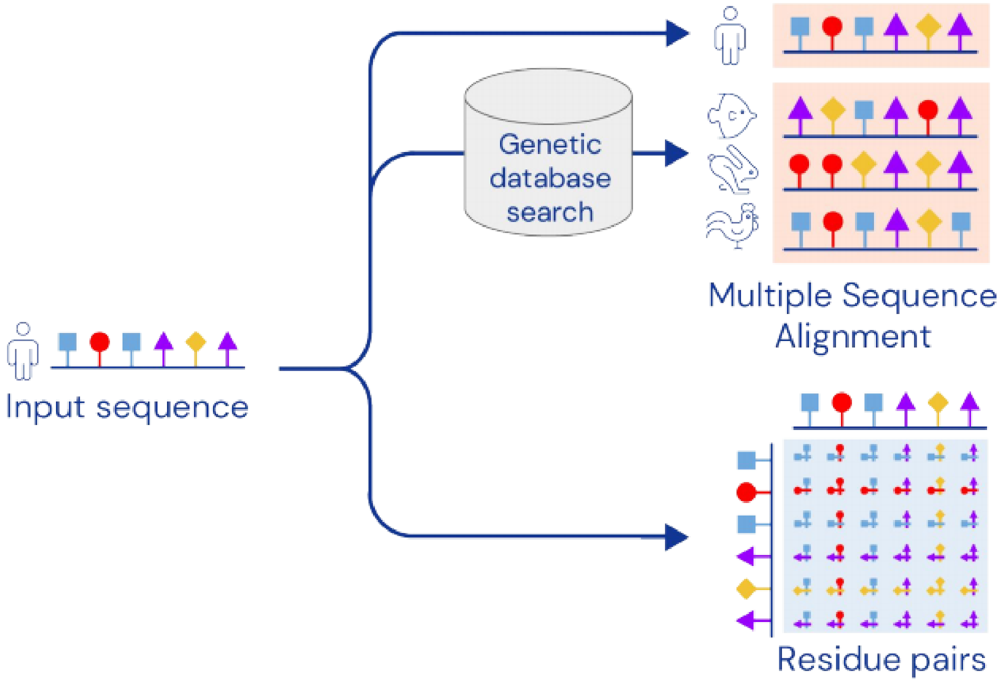
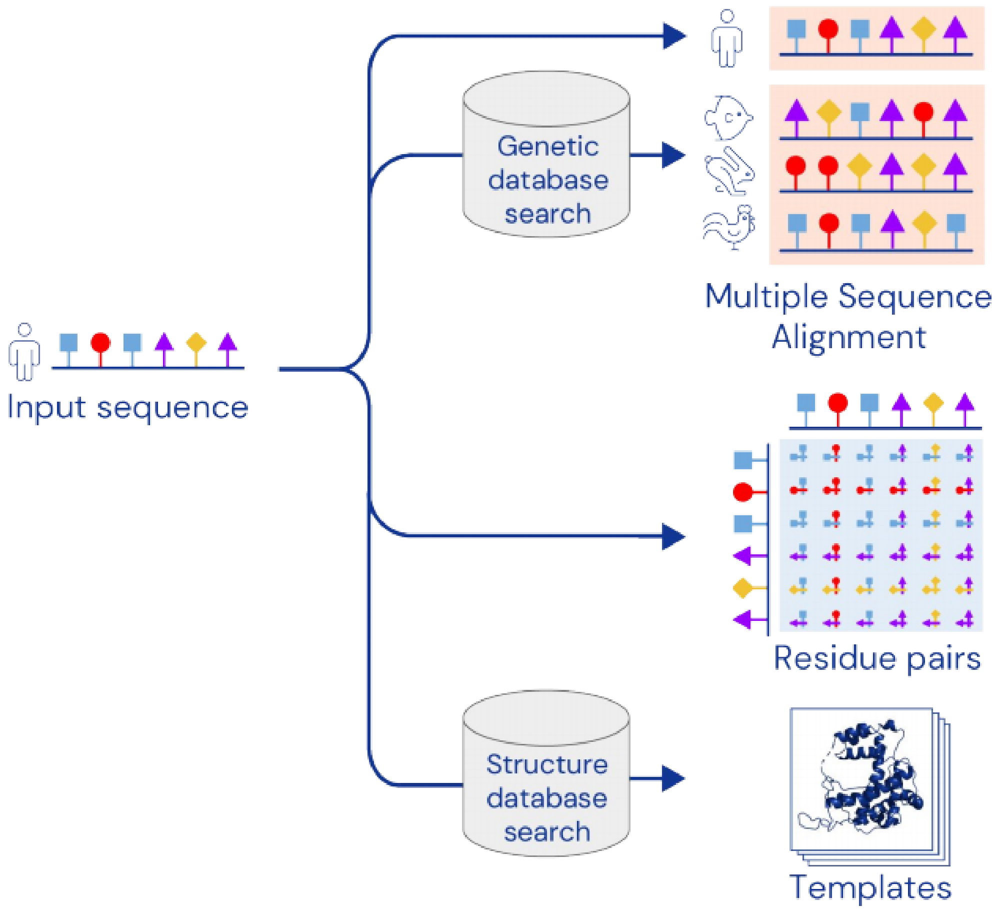
Inputs



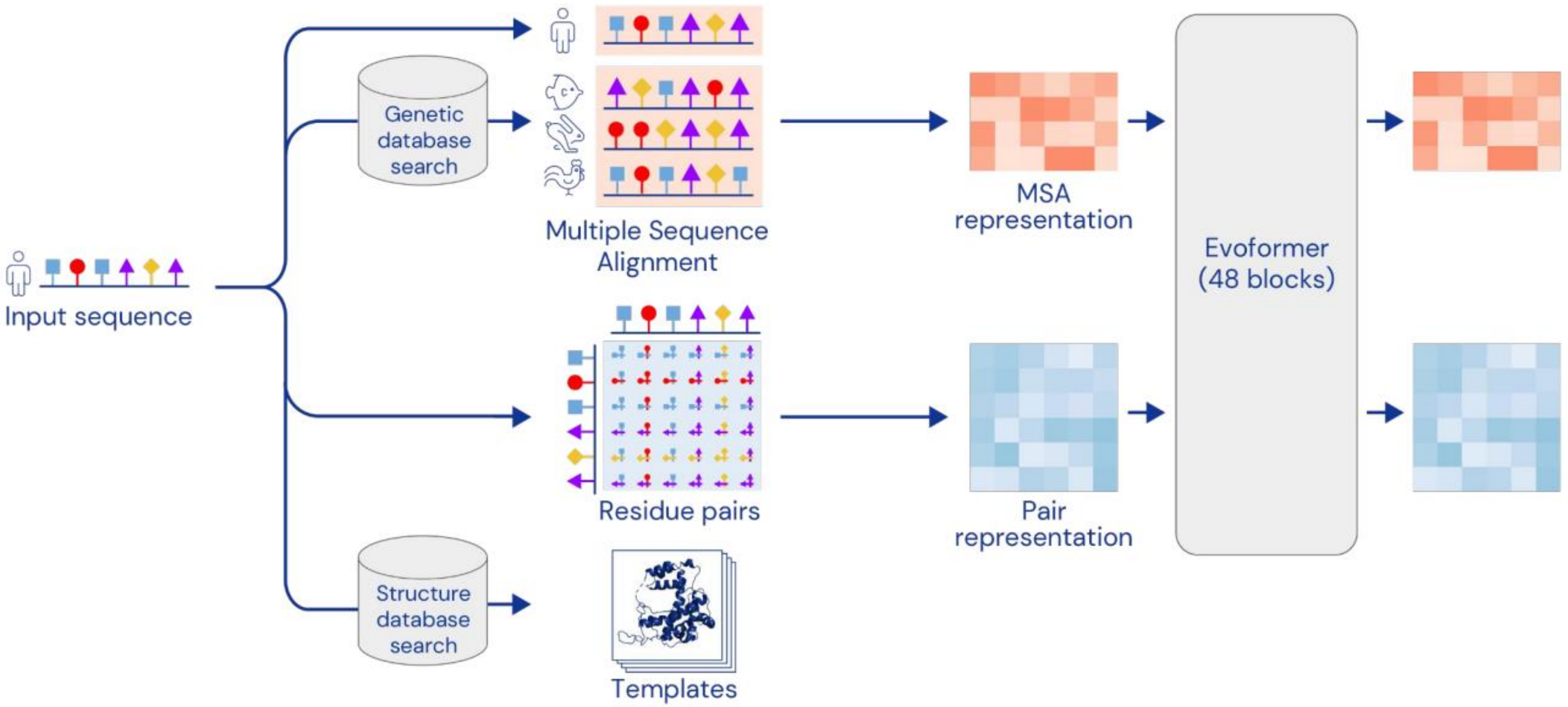
The input sequence is used to create an array of representations representing all residue pairs.

Inputs



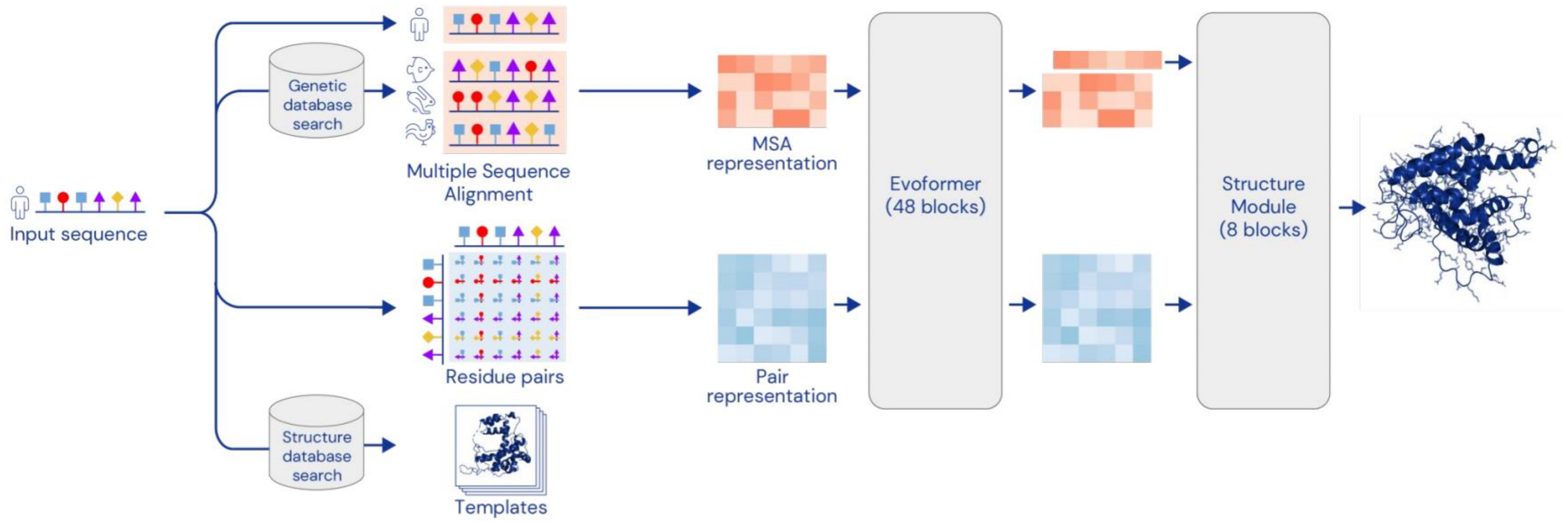
AlphaFold can also use template structures from the PDB, found using standard tools.
However, it often produces accurate predictions without a template.

Network



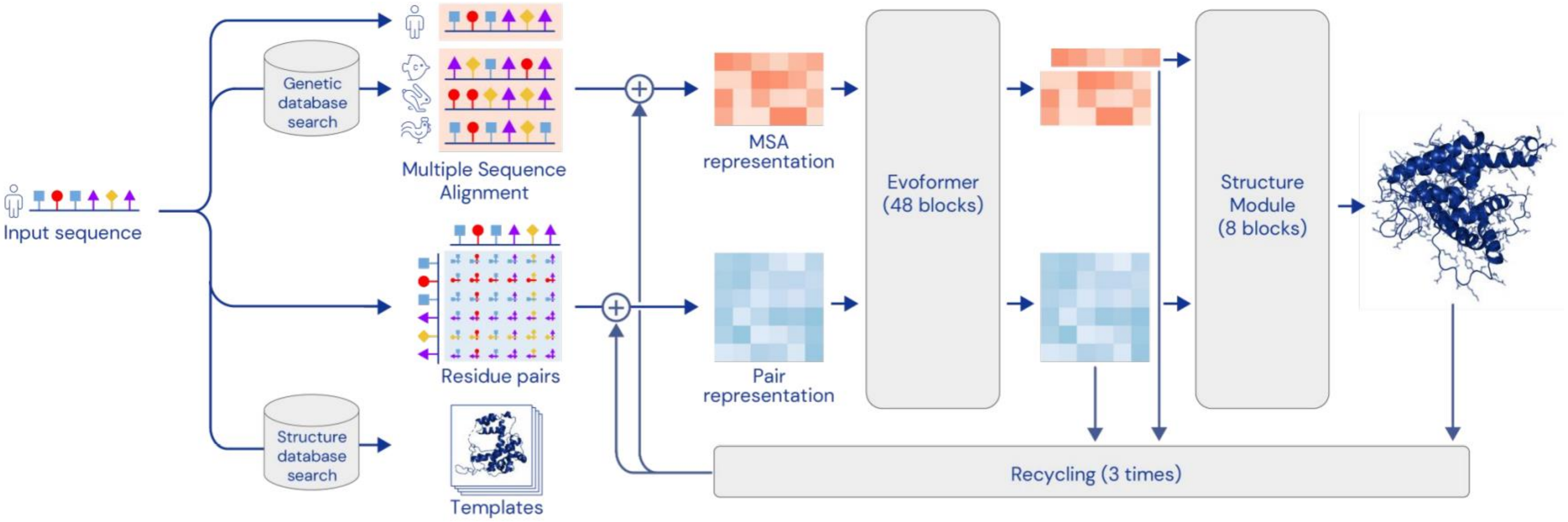
The Evoformer blocks extract information about the relationship between residues.
The MSA representation can update the pair representation and vice versa.

Network



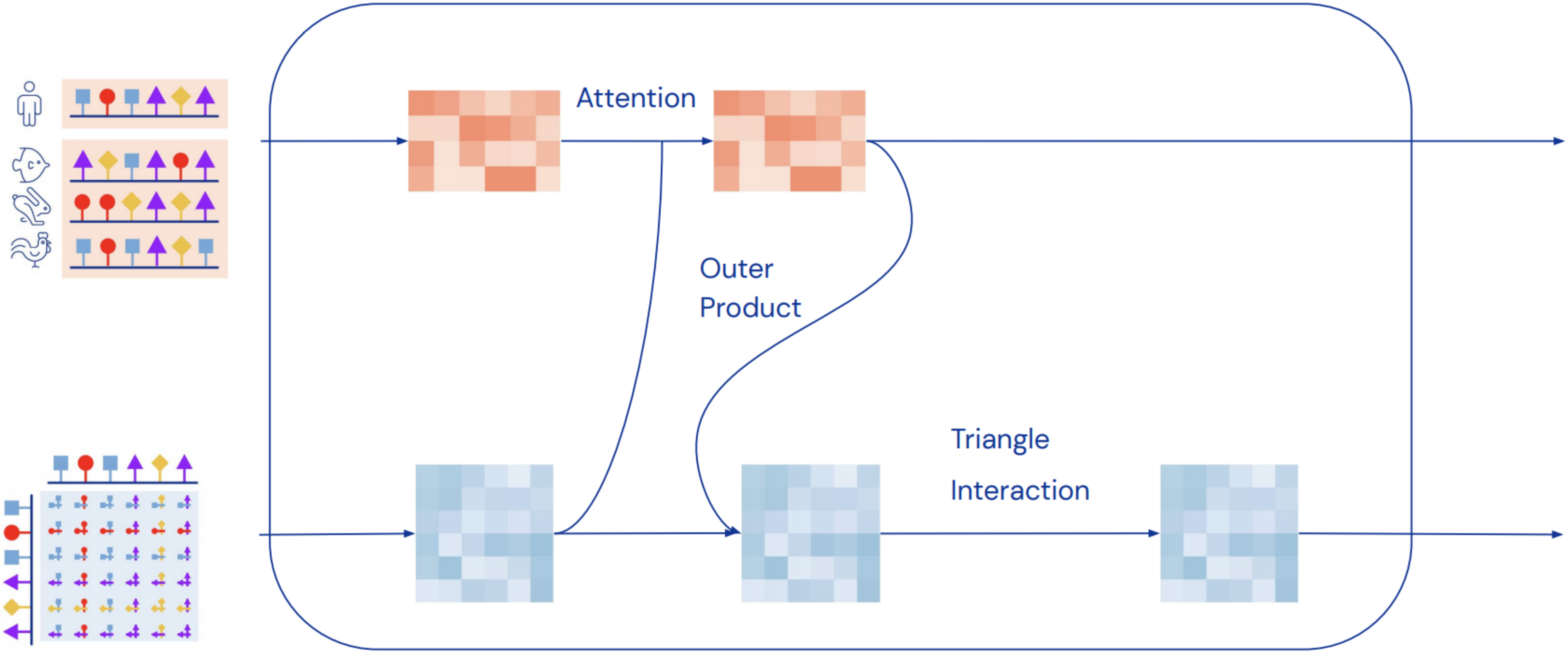
The Structure Module predicts a rotation + translation to place each residue.
A small network predicts side chain chi angles.

Network



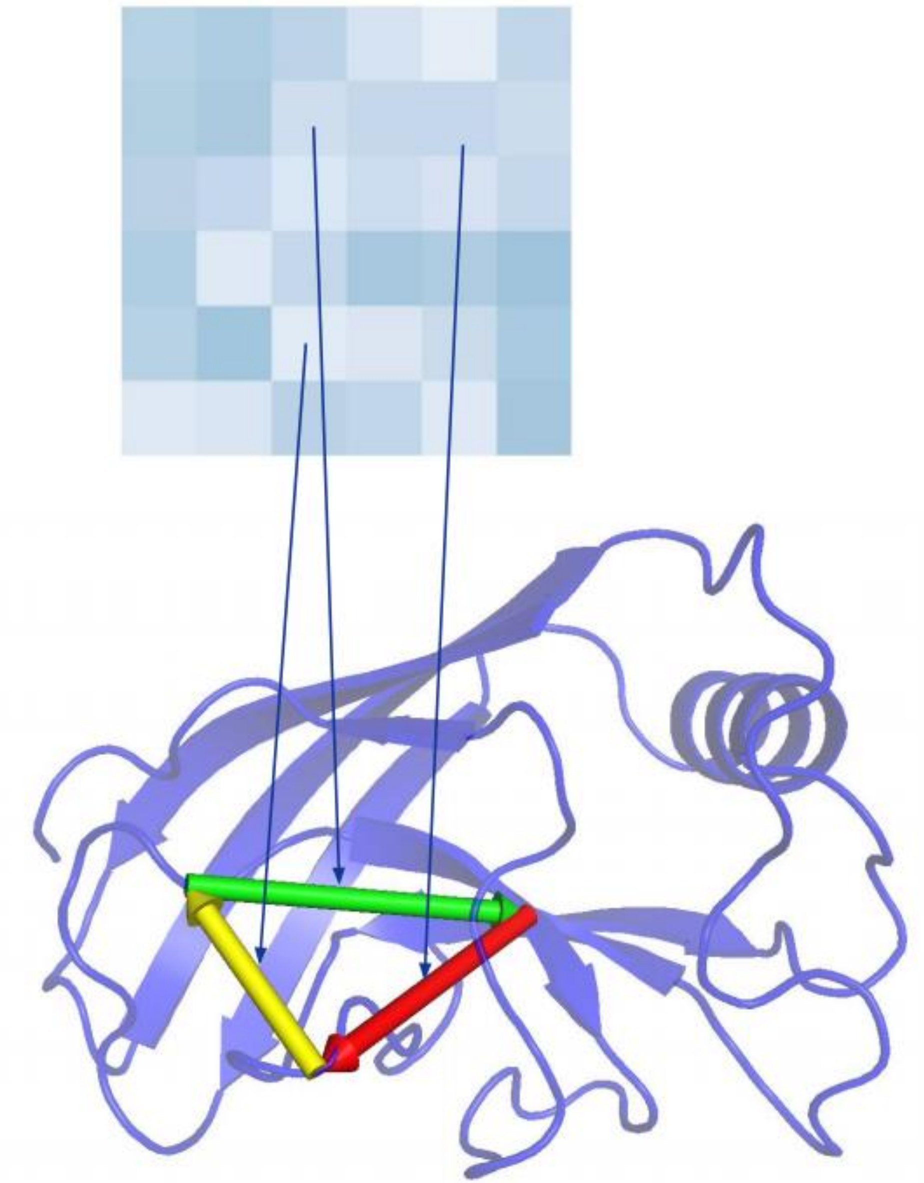
Feeding certain outputs back through the network again improves performance

Evoformer



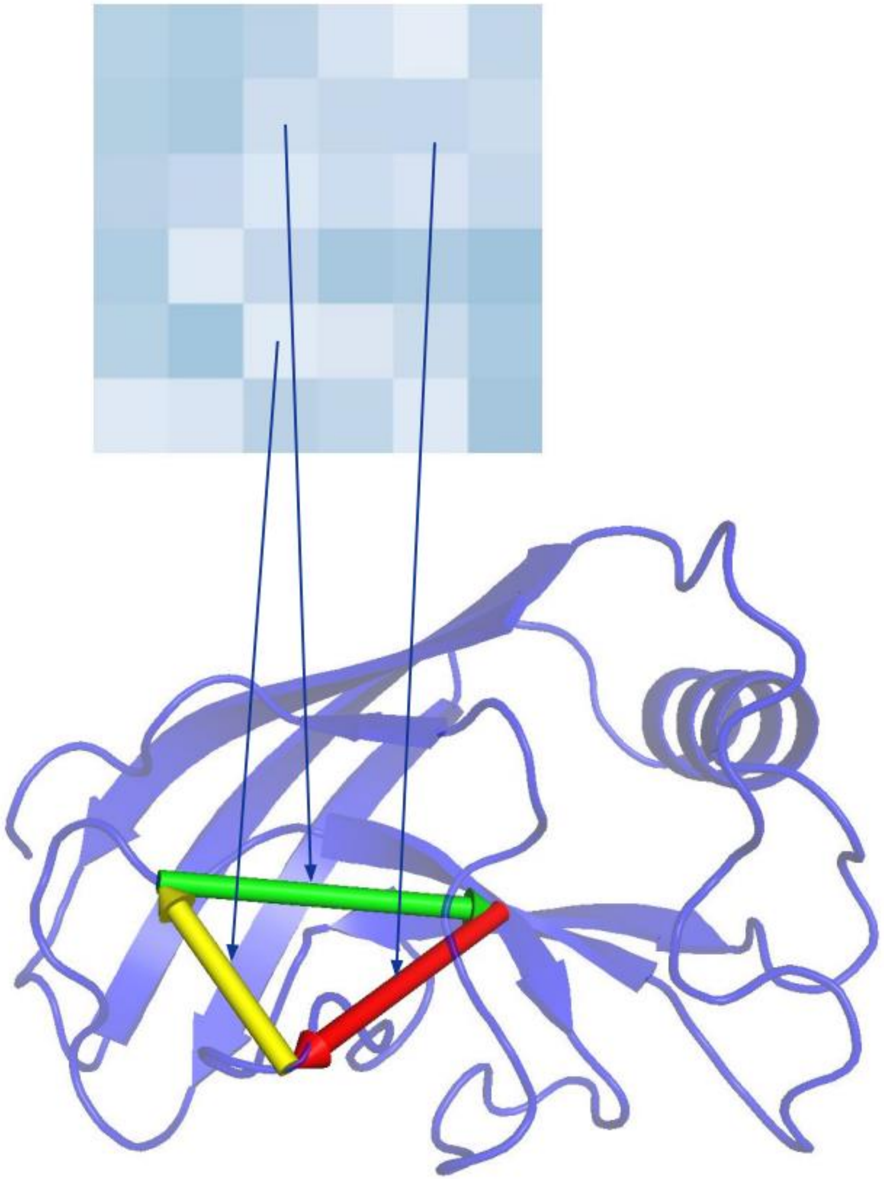
Triangle interaction

- ➔ **Take 3 points A, B, C**
  - ◆ If Distance AB and distance BC known strong constraint on AC (triangle inequality)
  - ◆ Evolution & Sequence gives information about relations between residues
- ➔ **Pair Embedding encodes relations**
  - ◆ Update for pair AC should depend on BC, AB



Graph inference

- **Graph**
  - Edges = Pairs of residues
  - We don't know the graph, need to infer it
  - Triplet relation in this language:
    length 3 cycles in graph
  - Update based on all cycles involving edge
- **Transitivity inductive bias**
  - This encodes transitivity bias of relations
    (E.g. triangle inequality, Loop Closure)



Evoformer

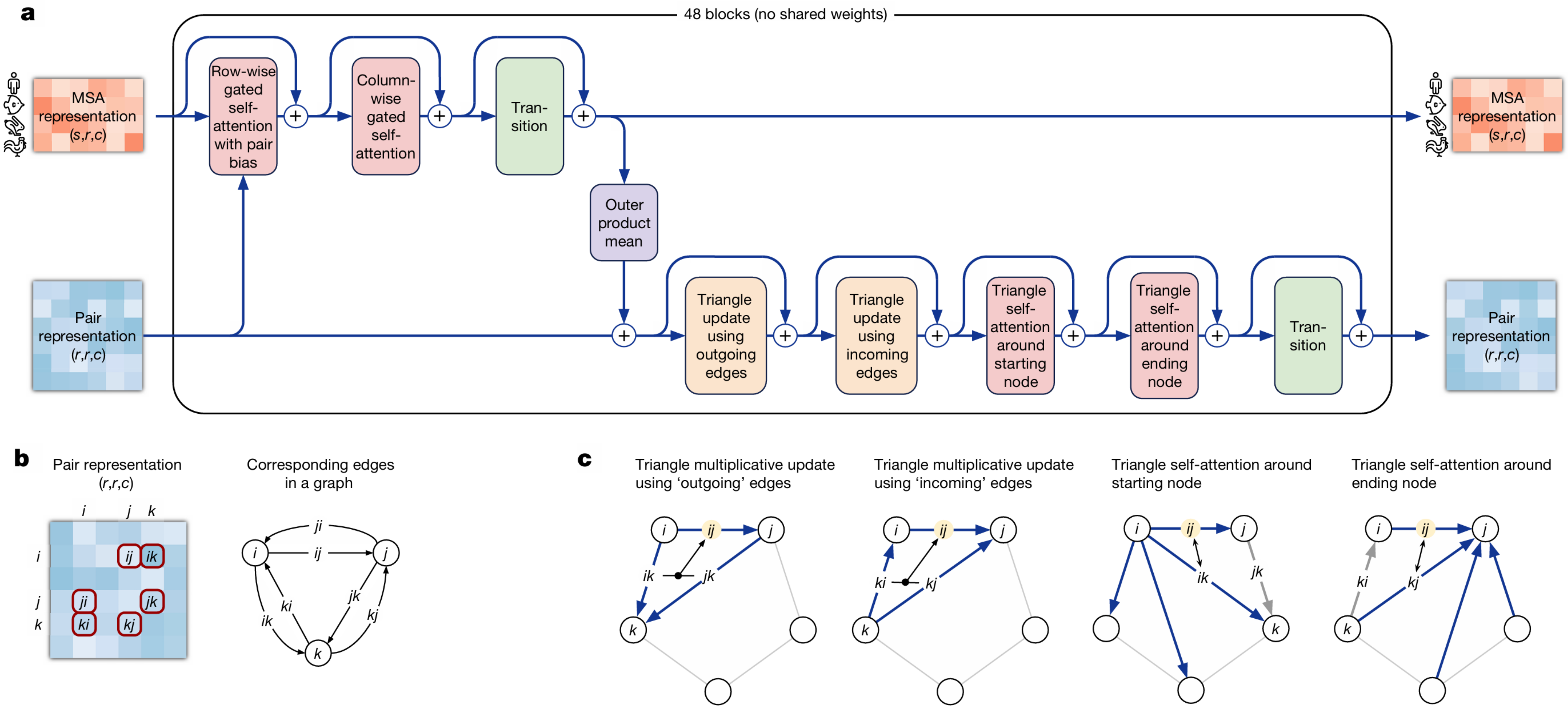


Fig. 3 of Jumper et al. Nature 2021 (the AF2 paper)

Network

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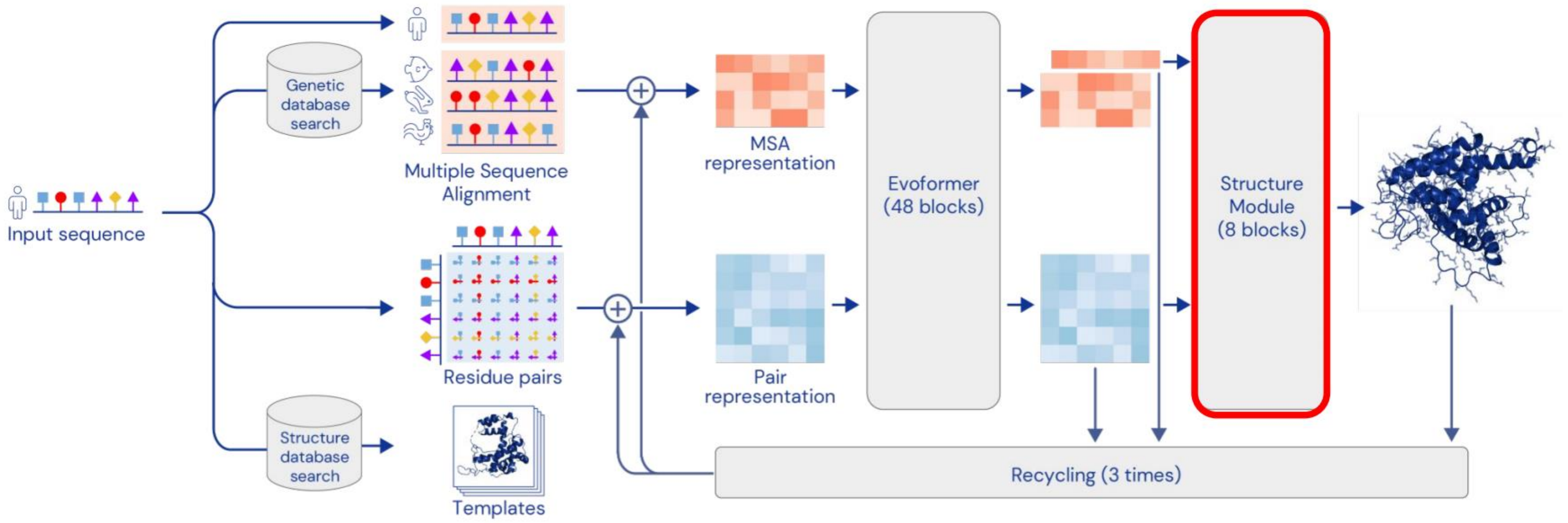
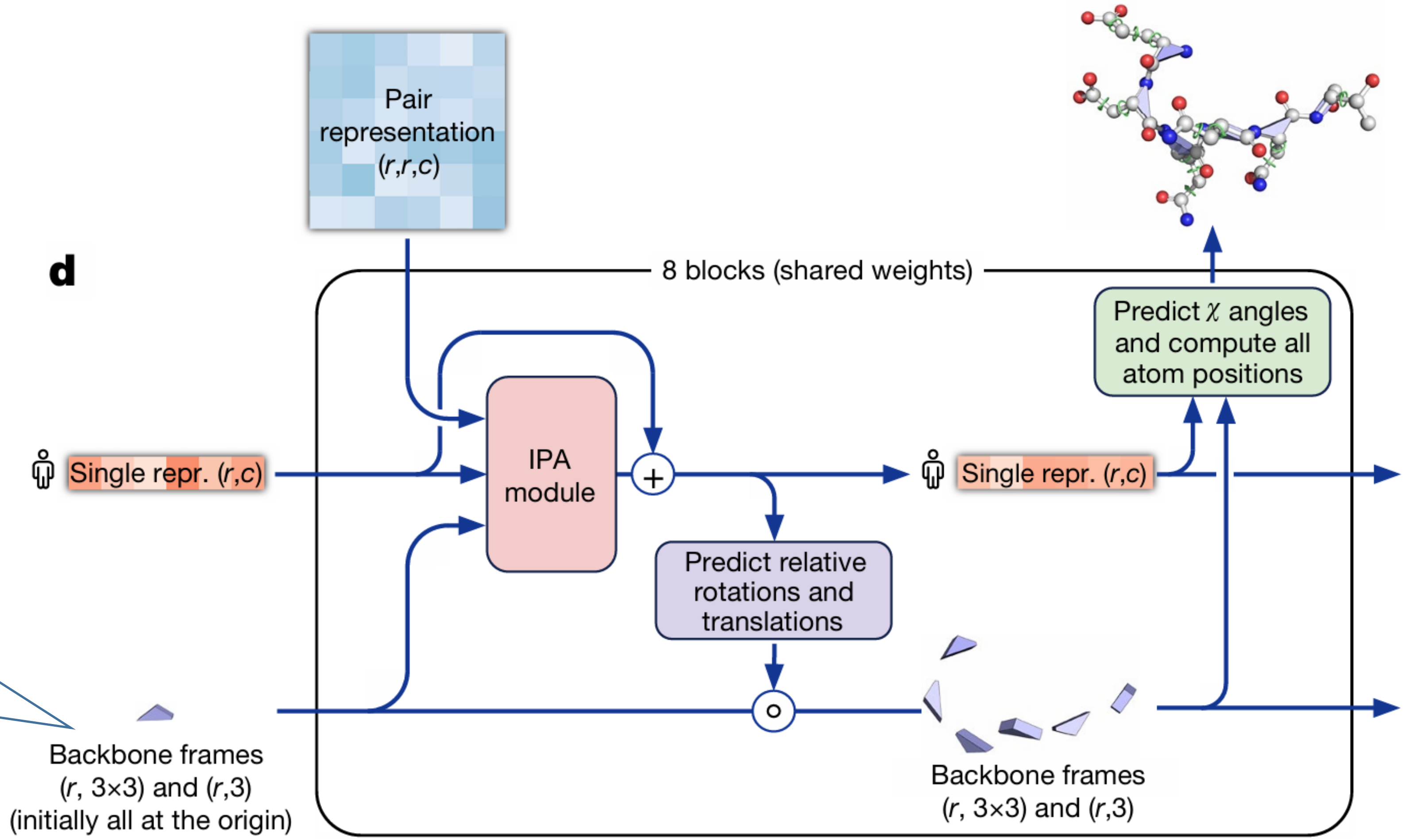


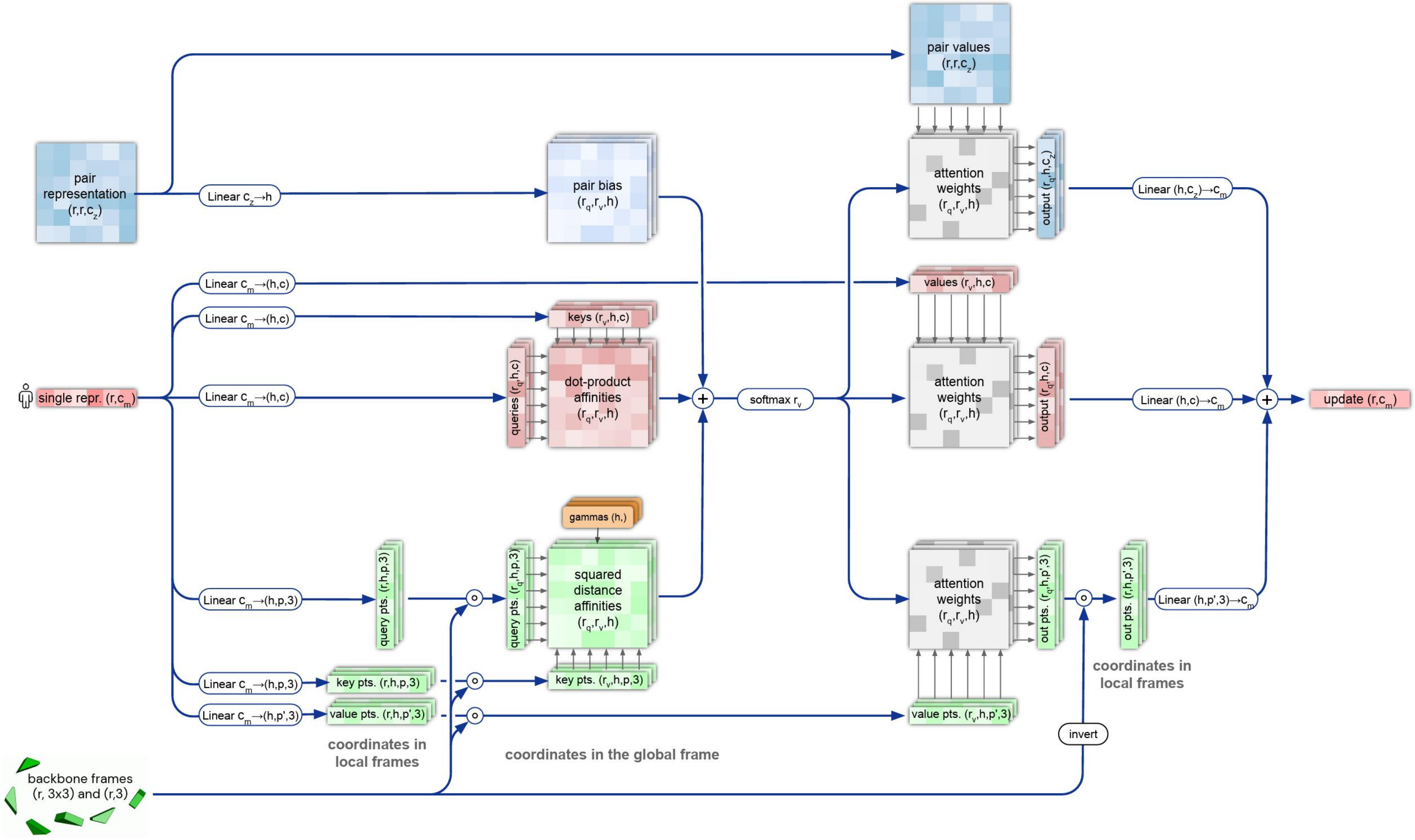
Structure module

The Structure Module maps the **abstract** representation of the protein structure (created by the Evoformer stack) to **concrete** 3D atom coordinates



"black hole initialization"

Invariant Point Attention (IPA) Module



Structure module

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- **End-to-end folding** instead of gradient descent
- Protein backbone = gas of 3-D rigid bodies (chain is learned!)

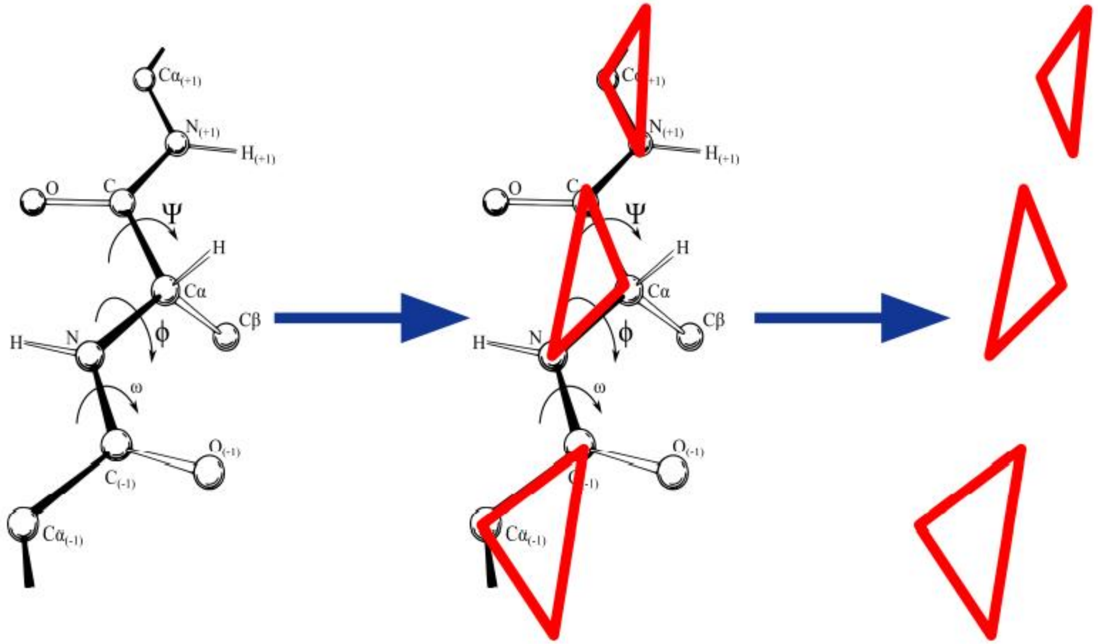
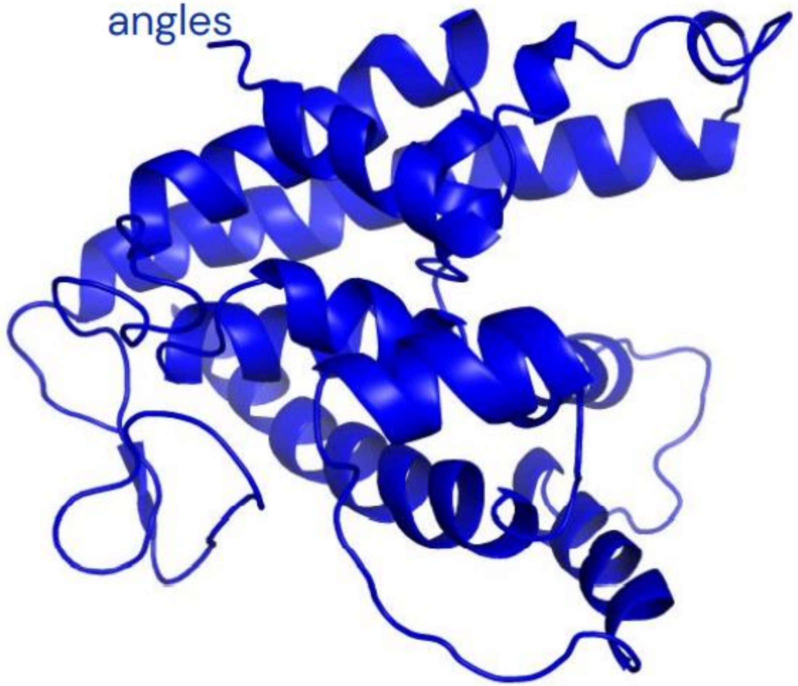


Image: Dcrjsr, vectorised Adam Rędzikowski (CC BY 3.0, Wikipedia)

- **3-D equivariant transformer architecture** updates the rigid bodies / backbone (Invariant Point Attention)
  - Also builds the side chains from torsion angles



Target: T1041

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Local frames

- Protein backbone = gas of 3-D rigid bodies (chain is learned!)
- Rigid body orientation defines local coordinate frame along chain

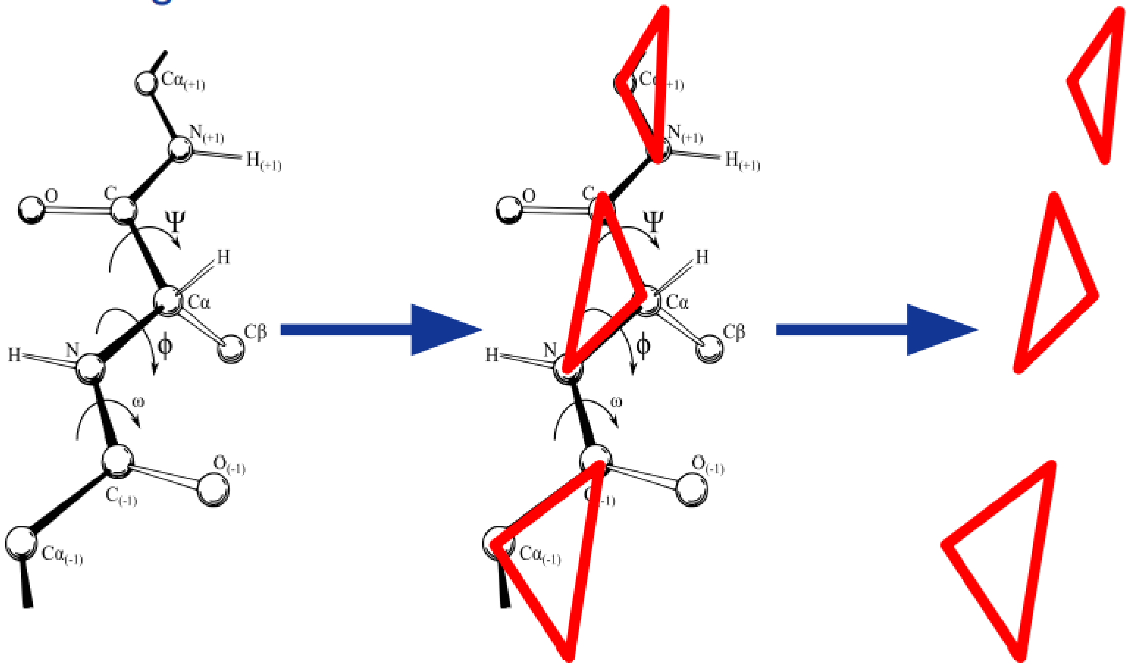
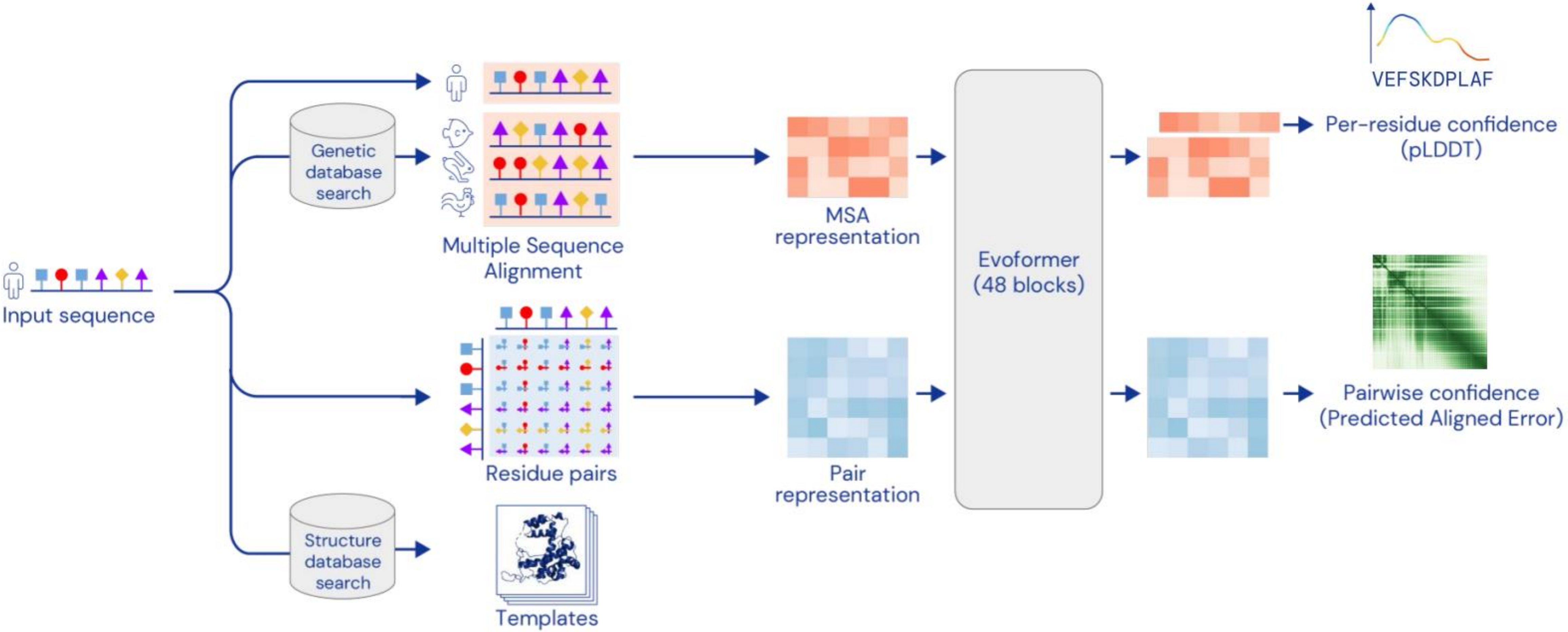


Image: Dcrjsr, vectorised Adam Rędzikowski (CC BY 3.0, Wikipedia)

- Coordinates measured in local Frame are **invariant to choice of global frame**

Confidence estimates



AlphaFold is trained to predict its confidence via simple supervision on the training set

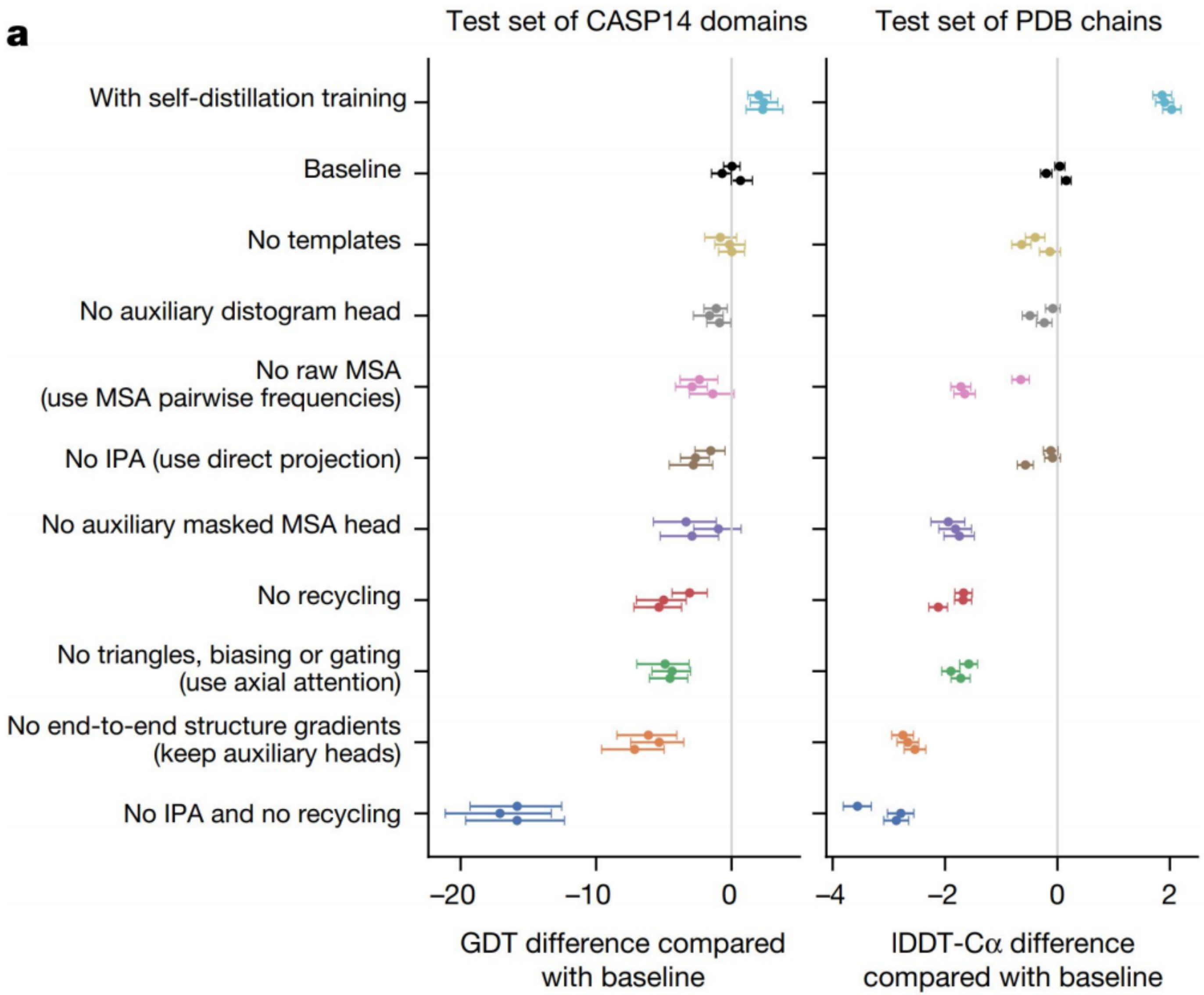
We use these confidences for ranking of predictions

All parts mattered

No single improvement is dominant

More important was the methodology of building the protein intuition into the model

Multiple ablations suggest strong interactions between many of the components



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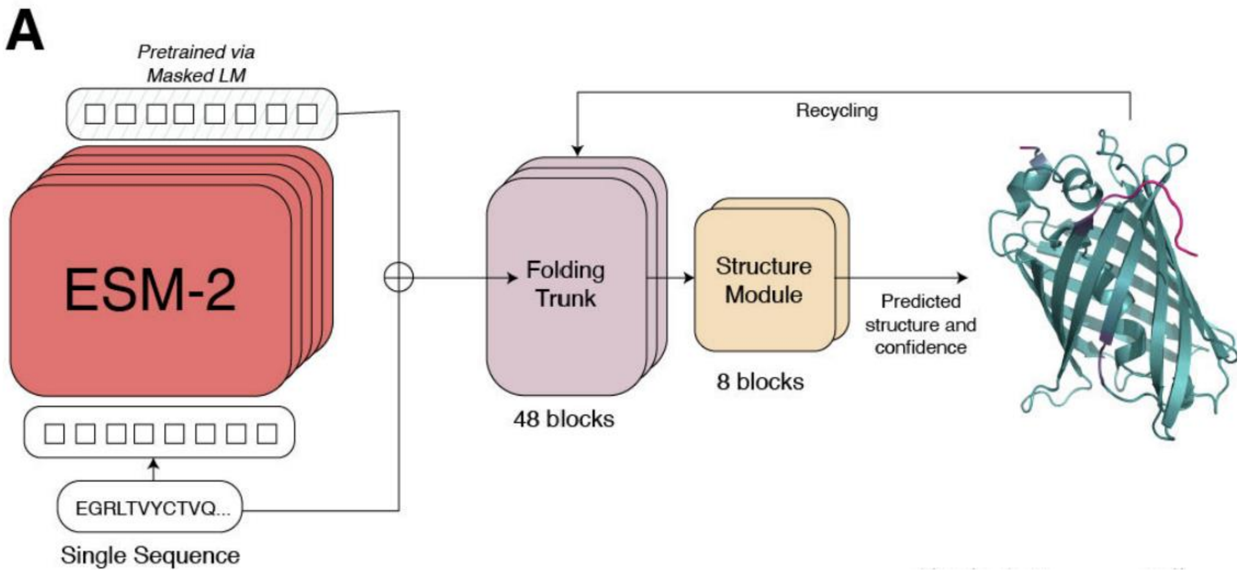
Some challenges of AF2

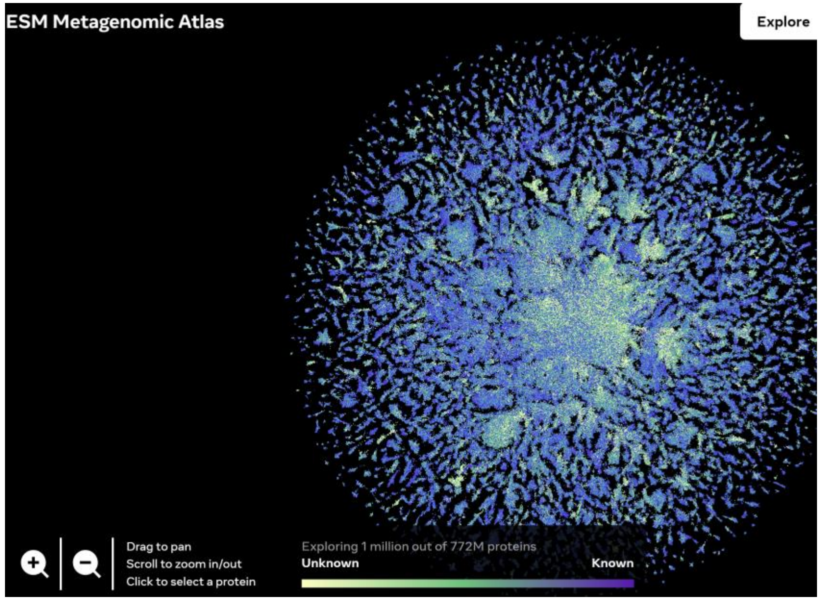
- **Extend AlphaFold2 to other major areas of structure prediction**
  - Increased multimer accuracy and completing the human structural interaction map
  - Structures of nucleic acid (e.g. mRNA) and ligand interactions
  - Conformational diversity
- **Structural metaproteome**
  - Extend the AlphaFold database to cover UniRef
  - Develop new tools to make these data useful
- **Prediction of mutational effects**
  - Changes in structure and stability
  - Binding affinity

AlphaFold2的protein language model (pLM)竞争者

- **AF2**巧妙地将**MSA**作为部分输入特征，并通过**Evoformer**来挖掘同源蛋白序列的共进化信号，但是缺点是搜同源序列需要较长时间，且基于MSA的模型对于序列突变不敏感
- 基于**单序列 pLM**的结构预测模型异军突起，包括OmegaFold，HelixFold-single，ESMFold



- Meta的 ESMFold，用ESM-2代替MSA，其余结构与AF2一致
- ESMFold的下游任务充分挖掘了预训练蛋白语言模型ESM-2的表达能力



基于ESMFold构建虚拟蛋白数据库ESM Metagenomic Atlas用于辅助药物研发

https://esmatlas.com/

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Diffusion in computational biology

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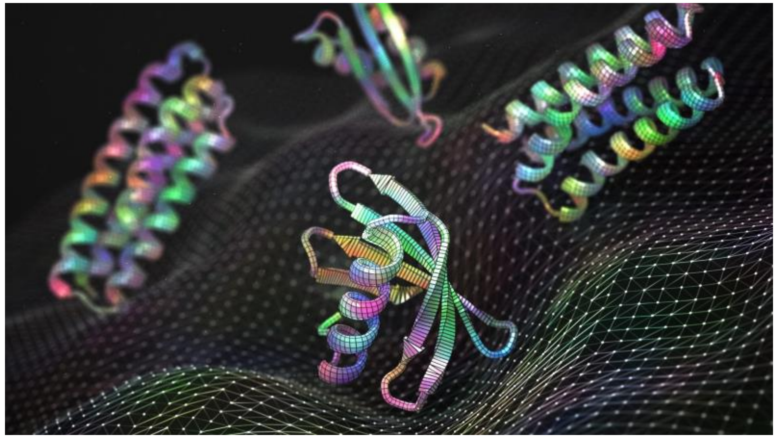
后AlphaFold2时代，研究者们努力发掘更大的AF2落地场景，即**从结构预测走向结构设计/生成**



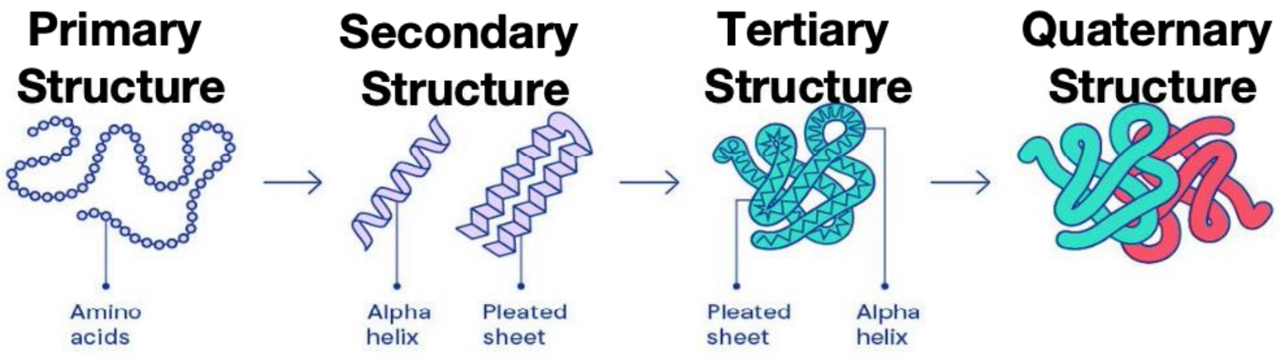
- 其中，扩散模型正是一个良好的赋能手段。2022-2023年也是扩散模型在计算生物学领域开始爆发的两年
- 主要集中于：**小分子生成**、**分子对接**、**蛋白质生成/蛋白质序列设计**等

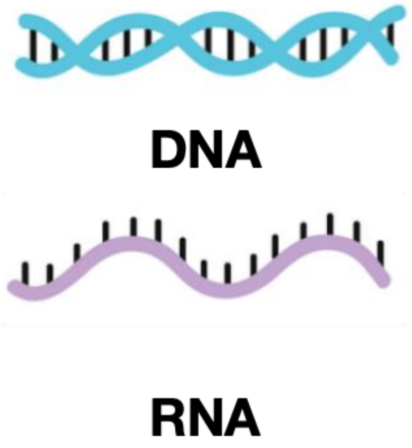
Tasks	Applications	Frame	Representative Methods
Molecule Modeling	Molecule Conformation Generation	SMLD	MDM [Huang *et al.*, 2022]
		DDPM	GeoDiff [Xu *et al.*, 2022], EDMs [Hoogeboom *et al.*, 2022], EEGSDE [Bao *et al.*, 2022], DiGress [Vignac *et al.*, 2022]
		SGM	Torsional Diffusion [Jing *et al.*, 2022], MOOD [Lee *et al.*, 2022], GDSS [Jo *et al.*, 2022], DGSM [Luo *et al.*, 2021a], DiffBridges [Wu *et al.*, 2022b]
	Molecular Docking	DDPM	FragDiff [Anonymous, 2023c], DiffLink [Igashov *et al.*, 2022], TargetDiff [Anonymous, 2023a], DiffBP [Lin *et al.*, 2022]
		SGM	DiffDock [Corso *et al.*, 2022]
Protein Modeling	Protein Generation	DDPM	SMCDiff [Trippe *et al.*, 2022], SiamDiff [Zhang *et al.*, 2022], DiffFold [Wu *et al.*, 2022a], ProSSDG [Anand and Achim, 2022], DiffAntigen [Luo *et al.*, 2022a], RFdiffusion [Watson *et al.*, 2022]
		SGM	ProteinSGM [Luo *et al.*, 2022a]
	Protein-ligand Complex Structure Prediction	DDPM	DiffEE [Nakata *et al.*, 2022]
		SGM	NeuralPLexer [Qiao *et al.*, 2022]

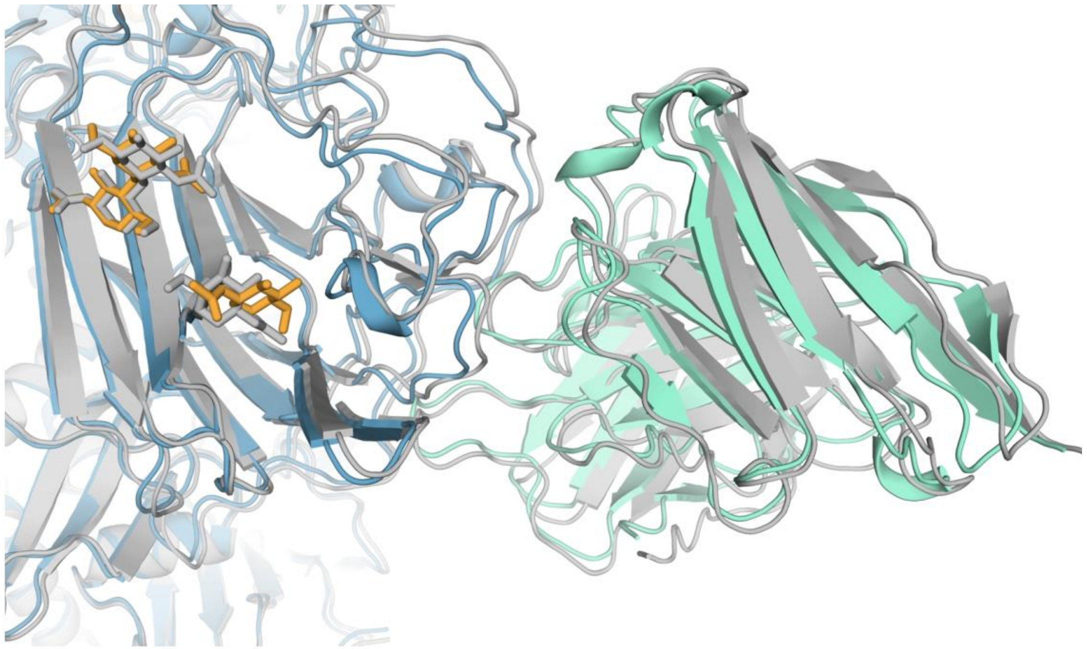
Fan, Wenqi, et al. *arXiv preprint* 2023.

AlphaFold 3

AlphaFold 3 predicts biomolecular structures





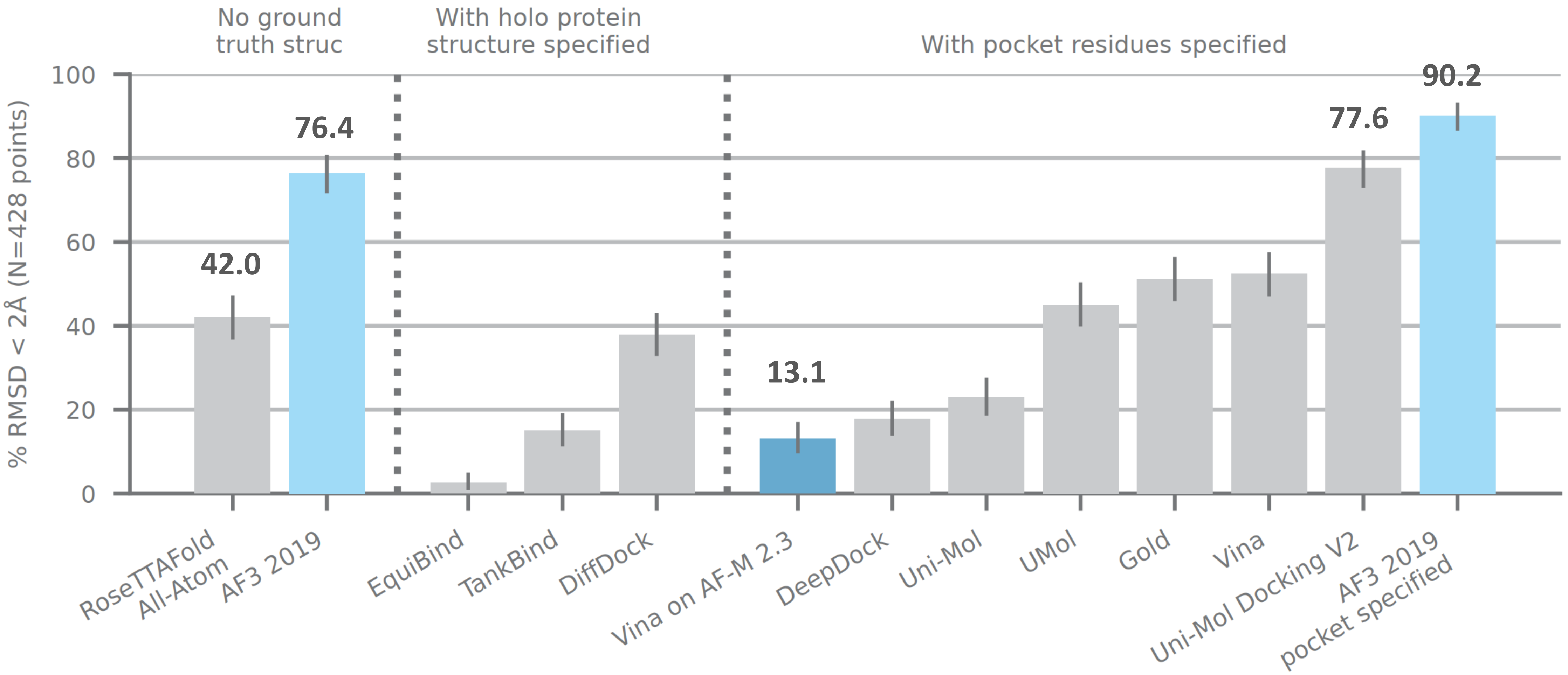


7R6R

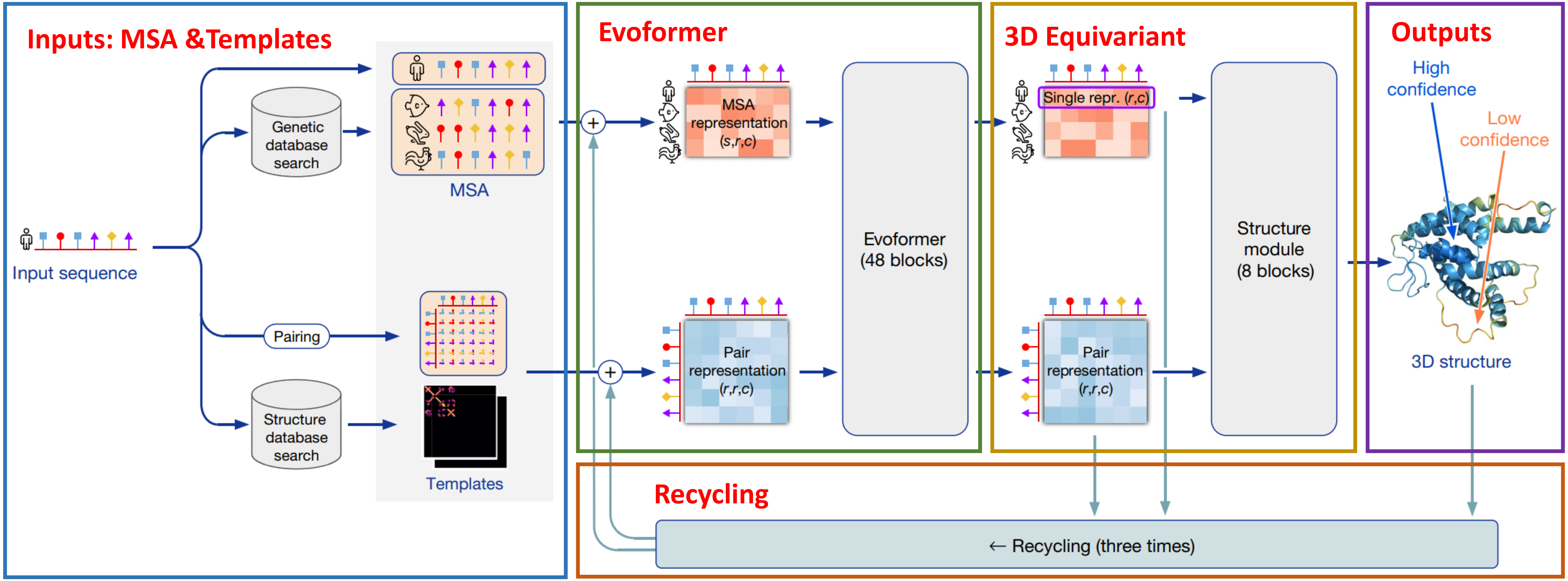
Ground truth shown in gray

AlphaFold 3 on ligand docking

PoseBusters Version 1 benchmark set (August 2023 release, 428 targets).



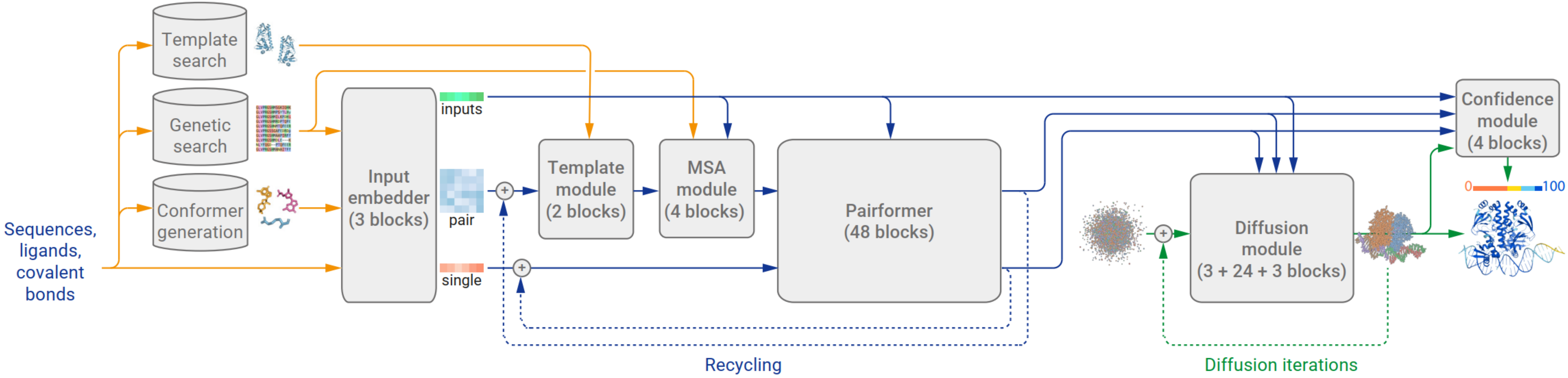
AlphaFold 2 (AF2)



- Evoformer (48 blocks) and Structure module (8 blocks)
- Recycling, self-distillation, IPA (Invariant Point Attention)

Jumper et al. Nature, 2021.

AlphaFold 3 (AF3)



AlphaFold 3 architecture for inference

Yellow: input data, blue: abstract network activations, green: output data

- **Pairformer**: replaces Evoformer, operates on pair representation and single representation
- **Diffusion module**: operates directly on raw atom coordinates
- **MSA processing**: pair weighted averaging, no column-wise attention

Abramson et al. Nature 2024.

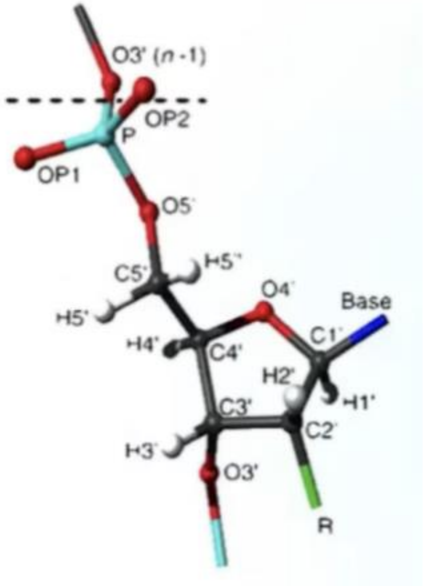
AF3 accommodate more molecule types

Tokenization

- A standard amino acid residue as a token
- A standard nucleotide residue as a token
- Modified amino acid / nucleotide residue is tokenized per-atom
- Ligands are tokenized per-atom

Token center

- Cα for standard amino acid
- C1’ for standard nucleotide
- The first and only atom of others



DNA atom numbering

Table 1 | Genetics databases for protein chains

Database	Search tool	Database-specific flags	Max sequences
UniRef90 [4]	jackhmmer [5]*	--seq_limit 100000	10,000
UniProt [6]	jackhmmer	--seq_limit 500000	50,000
Uniclust30 [7] + BFD [8]	HHBlits v3.0-beta.3 [9]		None
Reduced BFD [10]	jackhmmer	--seq_limit 50000	5,000
MGnify [11]	jackhmmer	--seq_limit 50000	5,000

Protein database, highly similar to AlphaFold2

Table 2 | Genetics databases for RNA chains

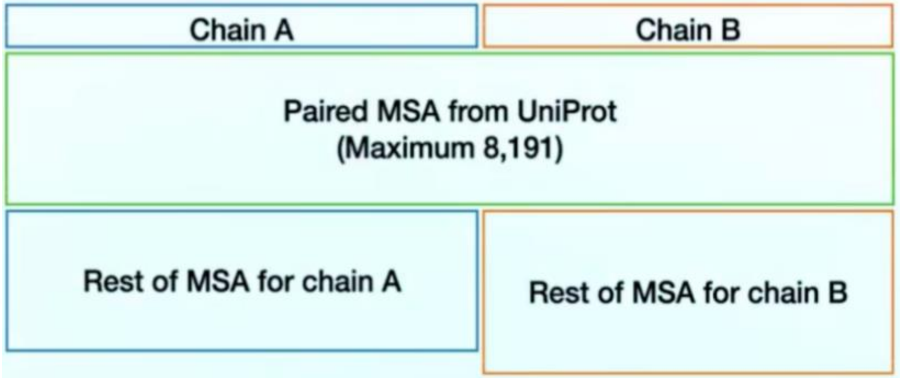
Database	Clustering tool	Search tool	Max sequences
Rfam [13]	mmseqs easy-cluster [14]	nhmmer [15]	10,000
RNACentral [16]	mmseqs easy-linclust	nhmmer	10,000
Nucleotide collection [17]	mmseqs easy-cluster	nhmmer	10,000

RNA database, highly similar to AlphaFold2

Genetic search during training

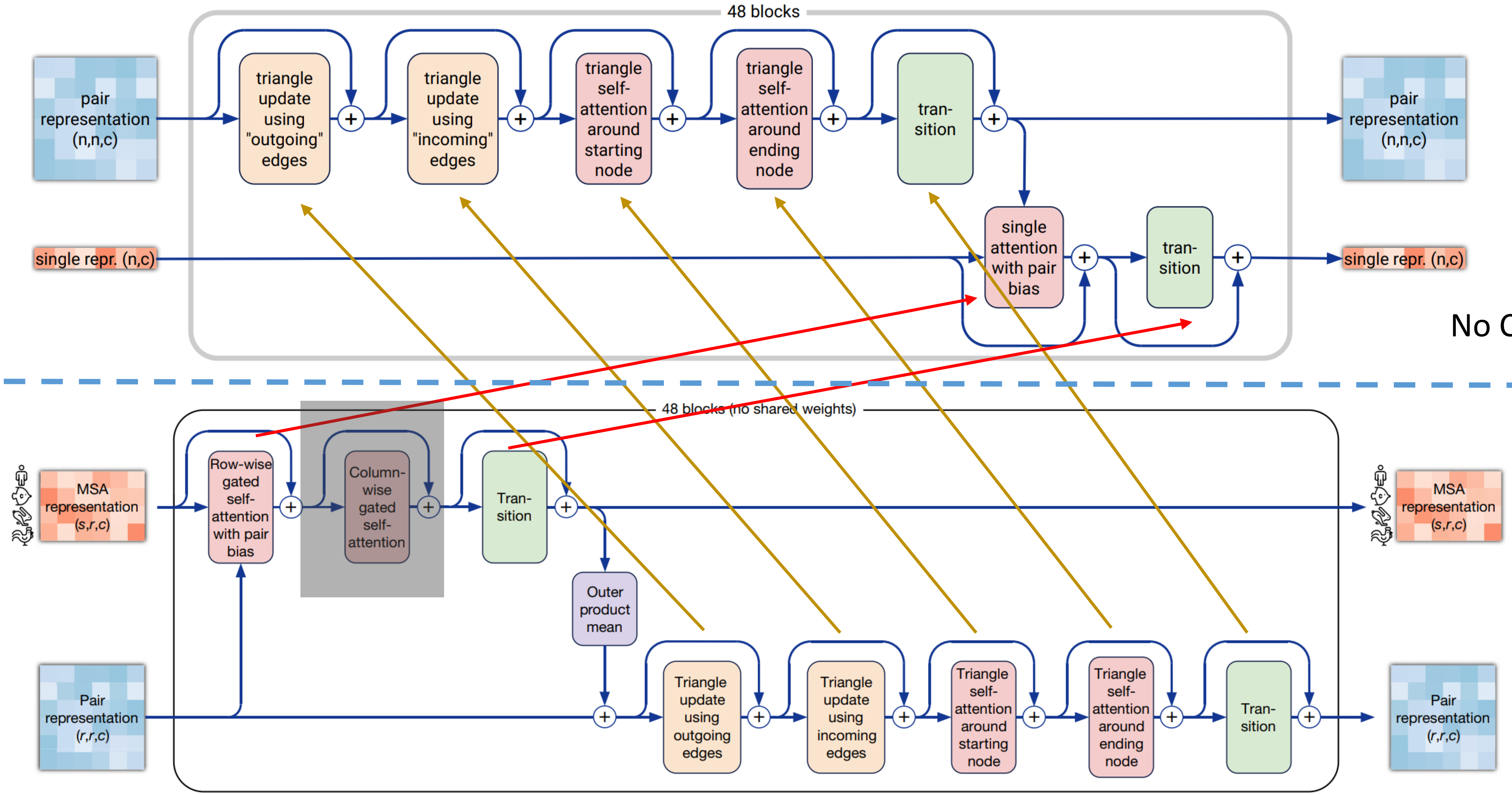
For training: all of them (BFD / reduced BFD)

For inference: reduced BFD



MSA processing (total maximum 16,384)

Pairformer vs. Evoformer

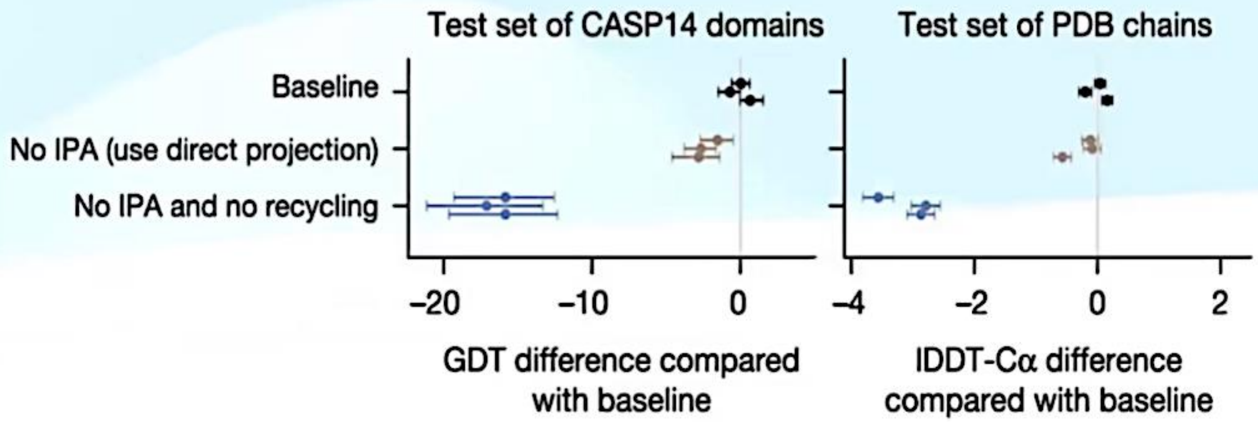


Pairformer (AF3)

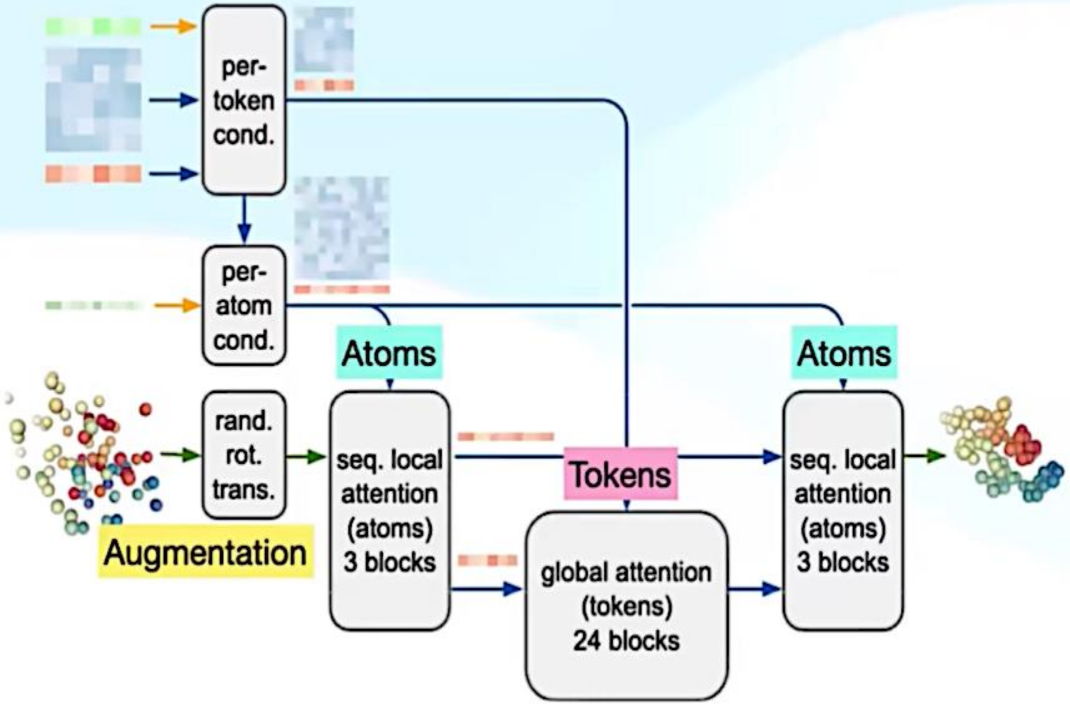
No Column-attention

Evoformer (AF2)

Structure Module in AF2 to Diffusion Modle in AF3



AlphaFold2 ablation studies: IPA is not necessary

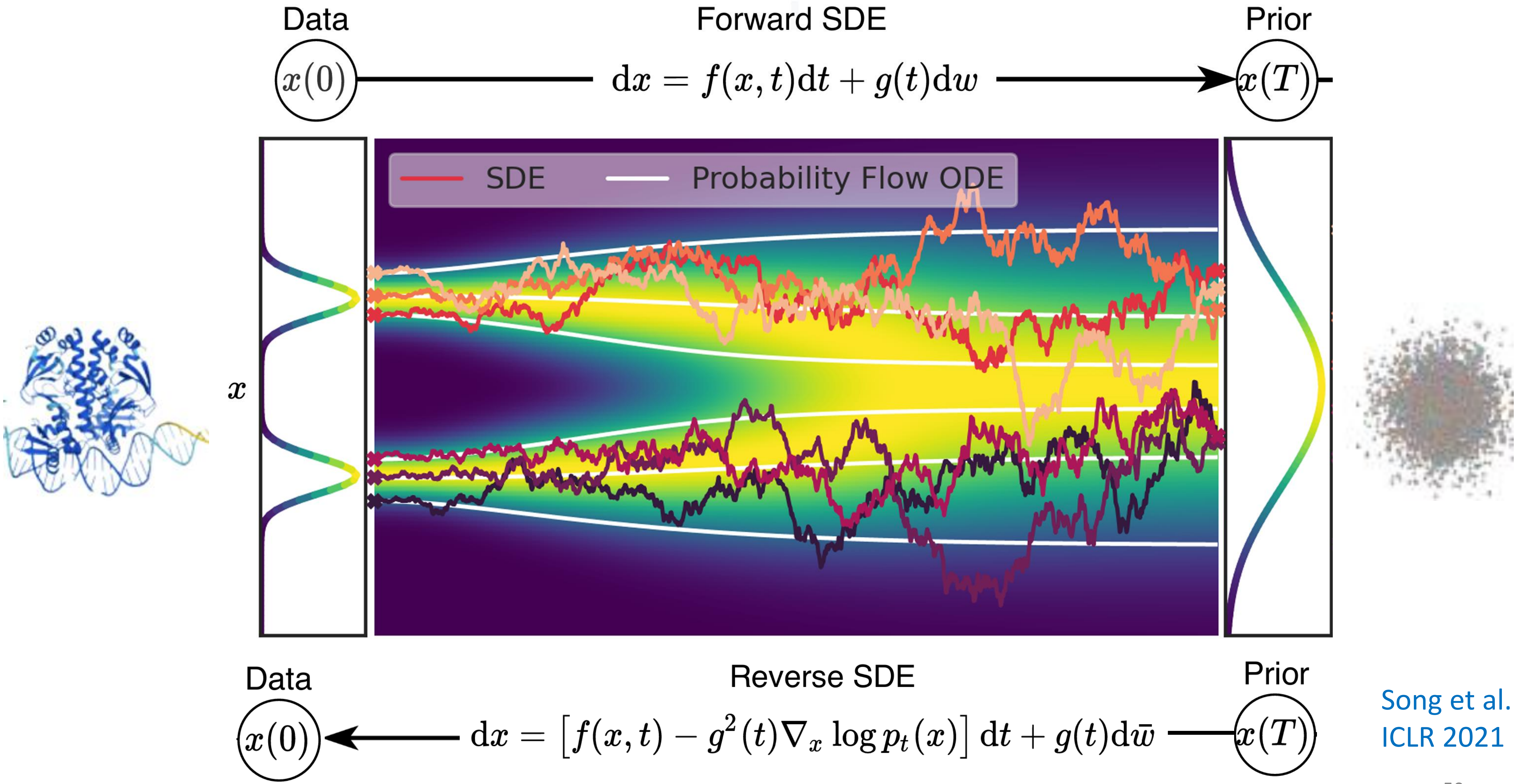


Diffusion Module

noised coordinates ➡ true coordinates

Only uses a linear layer to embed positions and and a linear layer project the updates, there are no geometric biases involved (e.g. SE(3) invariance)

Diffusion model used in AF3

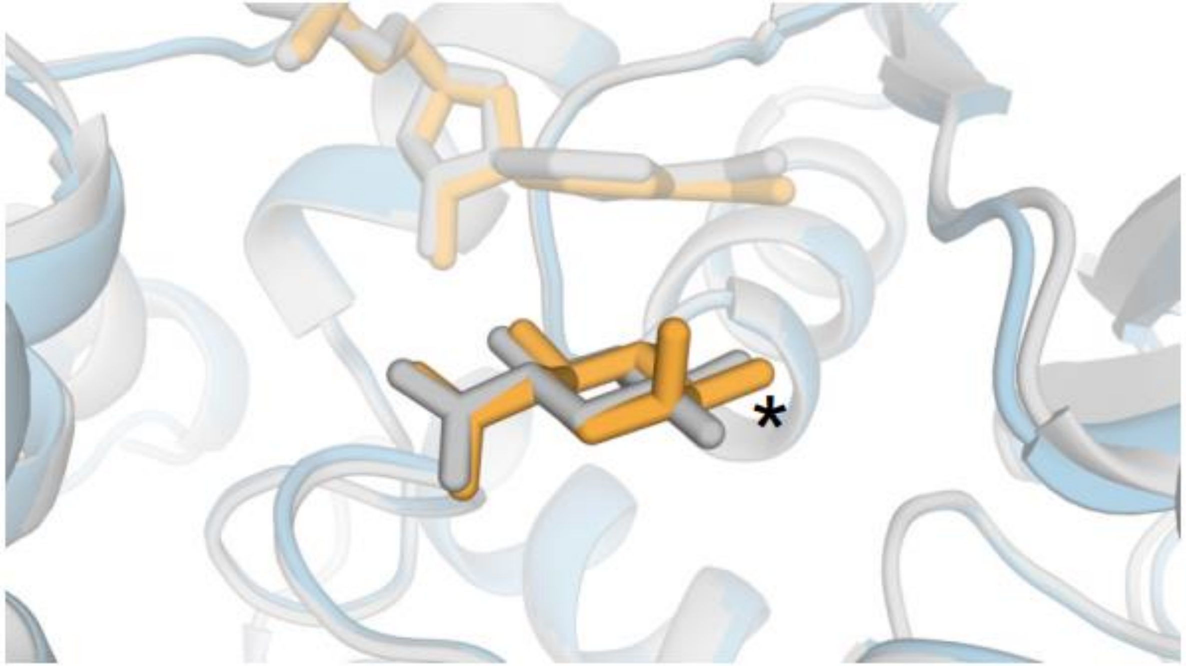


Song et al.
ICLR 2021

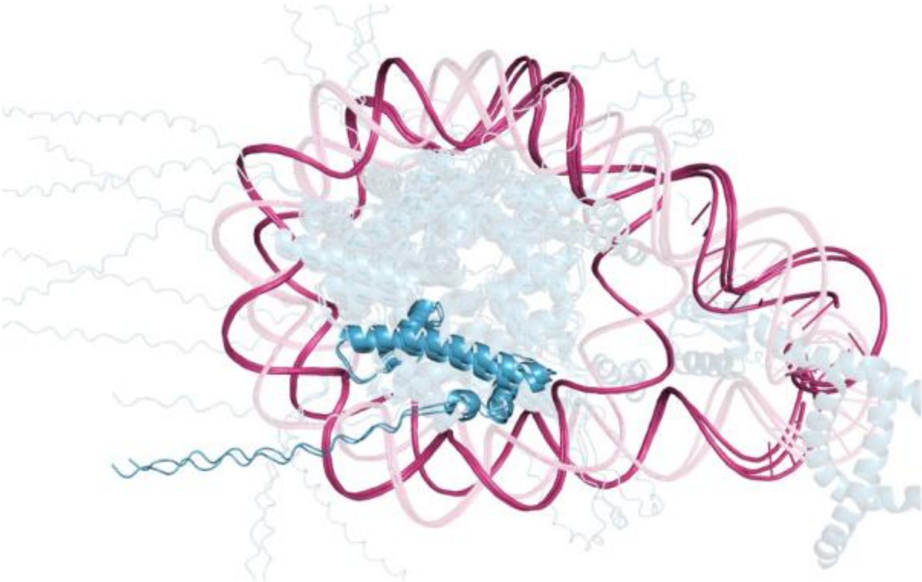
Diffusion model used in AF3

- **Diffusion process**
  - Adding noise to the data
  - Denoising to retrieve original data
- **Conditional diffusion model**
  - Generates data based on given conditions
  - Sequence representation as the condition (pair & single representations) in AF3
- **Diffusion module in AF3**
  - Multi-scale learning:
    - Small noise emphasizes local stereochemistry
    - High noise emphasizes large-scale structure
  - Generative process:
    - Produces a distribution of answers
    - Sharply defines local structures despite uncertainty

Limitations of AF3

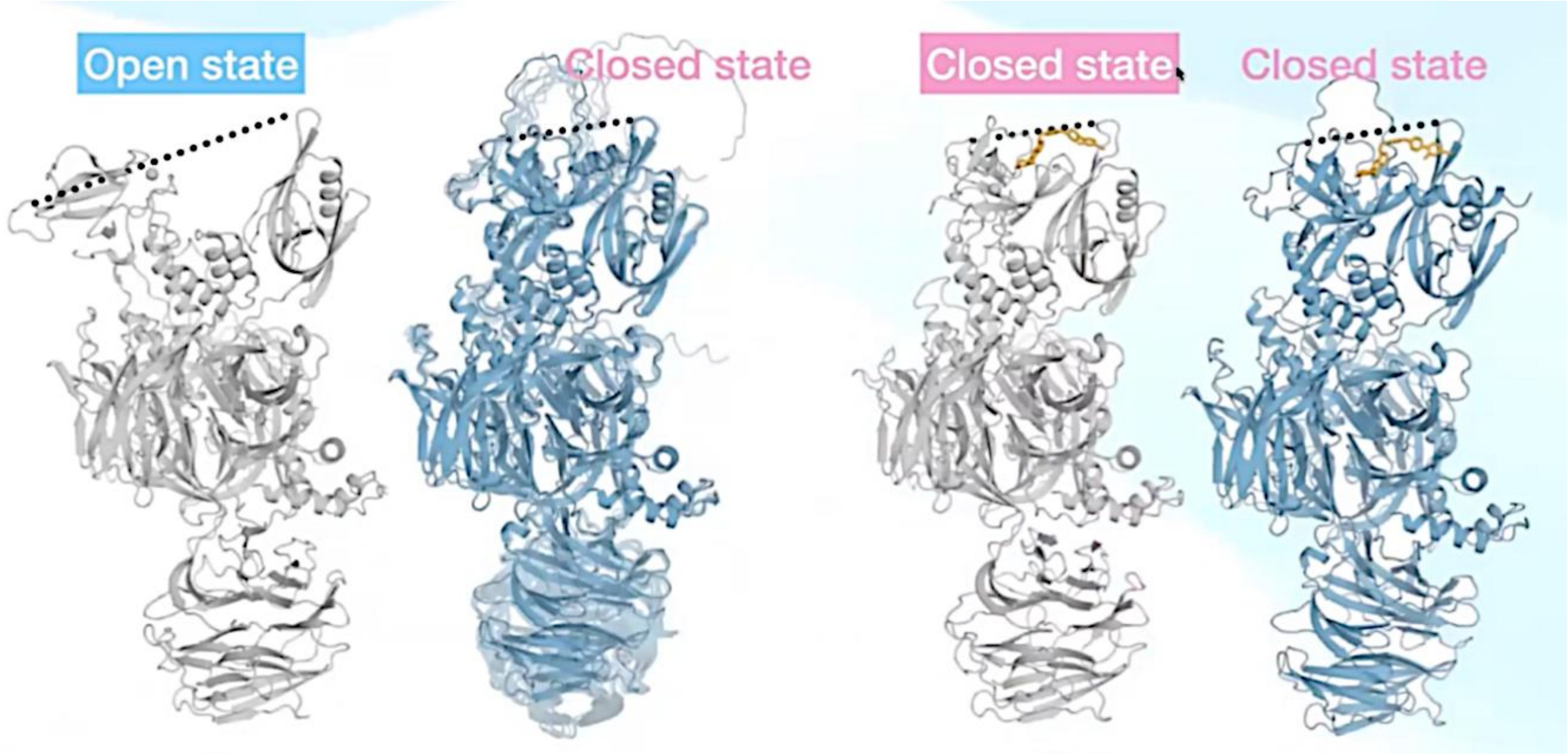


Incorrect chirality
PDB: 7CTM, incorrect chiral centre indicated by *



Clash / overlapping atoms
PDB: 7PEU, highlighted are overlapping protein chainsand self-overlapping DNA chain

Limitations of AF3



Limited conformation coverage predicted by AlphaFold 3 of E3 ubiquitin ligases

- Grey: ground truth of cereblon (PDB 8CVP apo, PDB 8D7U holo)
- Blue (left): AF3 prediction of Apo structure (10 samples)
- Blue (right): AF3 prediction of Holo structure (10 samples)
- Dashed line indicates distance between N- and C-terminal domain

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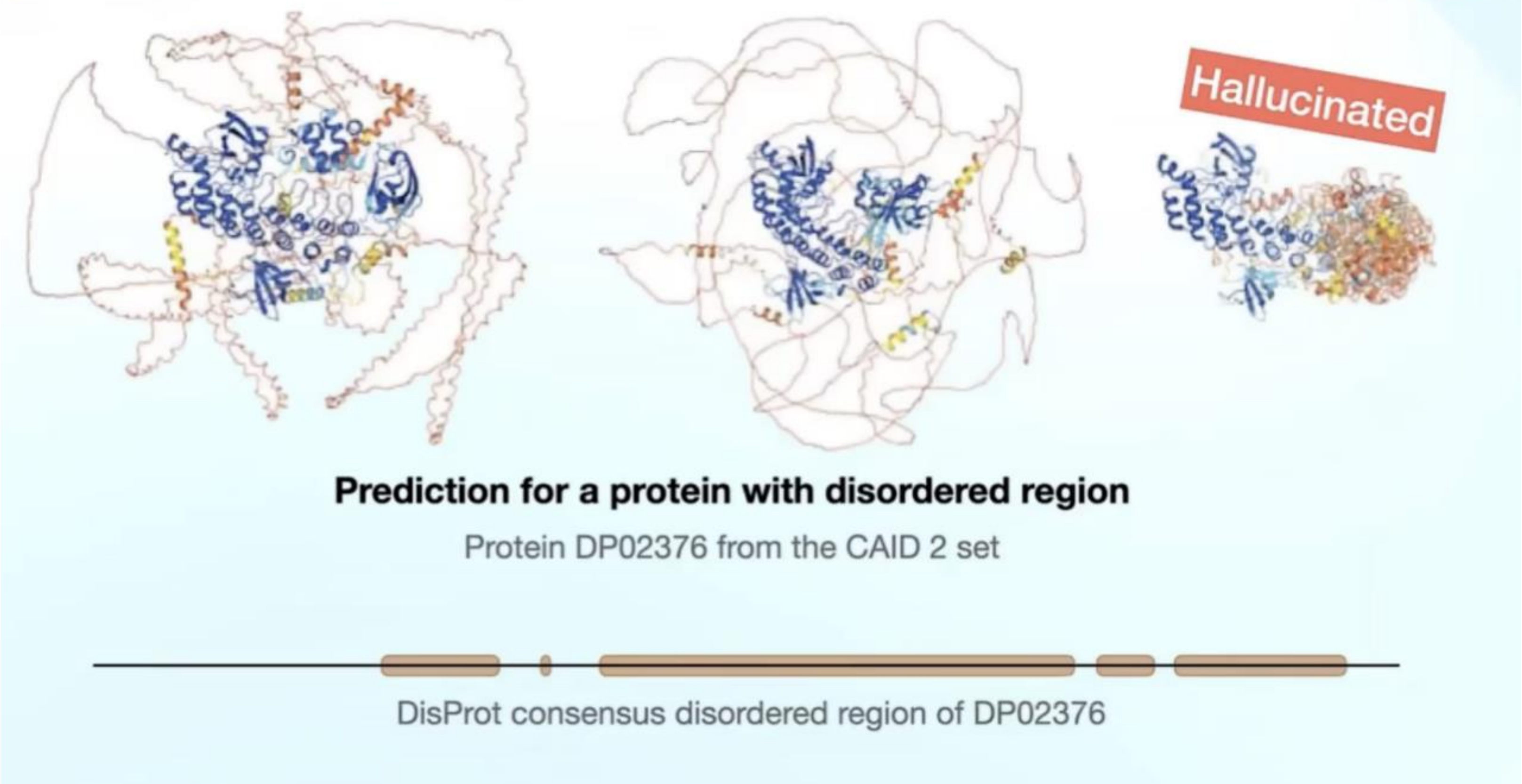
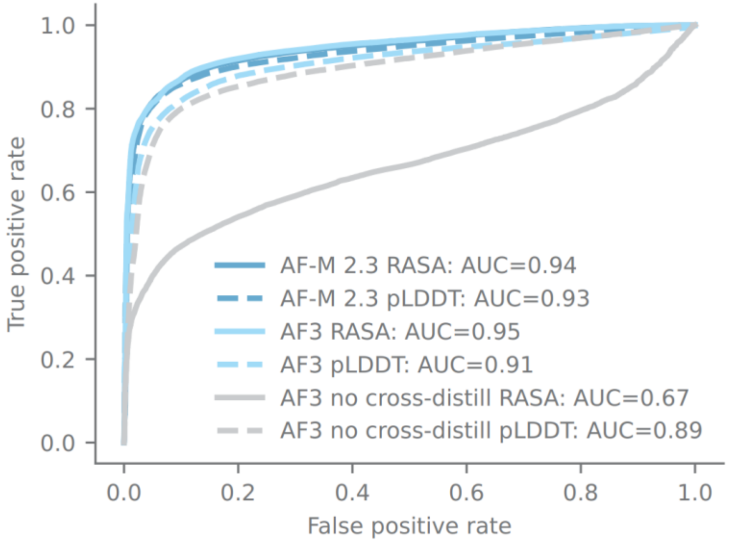
Hallucination and cross-distillation

AlphaFold-Multimer v2.3　**AlphaFold3**　**AF3 w/o cross-distillation**





Predict disorder residues in the CAID 2 set

- RASA (relative accessible surface area)
- pLDDT

How does AF3 differ from AF2?

上海科技大学 ShanghaiTech University

- **Extended capabilities:**
  - AF3 models general molecules, not just proteins
  - Allows specifying ligands and modified residues
- **Structure representation:**
  - AF2 used rigid body frames for amino acids
  - AF3 models as independent atoms with global coordinates
- **Computational efficiency:**
  - AF3 groups atoms into tokens for efficient computation
  - AF3 uses minimal spatial inductive bias
- **Generative diffusion head:**
  - AF3 adds Gaussian noise to atom coordinates and removes it using MSE loss
  - AF3 only applies bond loss for bonded ligands and glycans

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How does AF3 differ from AF2?

- **Simplified architecture:**
  - Introduced "Pairformer" architecture
  - Removed MSA representation and column-wise attention
  - Replaced Evoformer stack with Pairformer stack
- **MSA handling:**
  - Dense presentation of unpaired MSA sequences, allowing more MSA sequences per-chain
- **Activation function:**
  - Replaced ReLU with SwiGLU for better results
- **Improved datasets:**
  - New distillation sets, re-balanced at the interface level
  - Removed restrictions on pLDDT (80) and protein length (200) for monomer distillation set

Recap and discussion

- The AlphaFold (series) is by far the most successful case of AI for Science
  - AF2 is mainly based on Transformer
  - AF3 still uses Transformer, but has simplified some parts of AF2 (e.g. replacing Evoformer with Pairformer), and AF3 relies on Diffusion model
- Questions for discussion:
  - Why is AlphaFold so successful?
  - What are the limitations of AF2 and AF3? Will some of them be solved soon in the next few years?
  - What are pitfalls and limitations of generative AI techniques like Diffusion models? And how to address them?
  - Is scientific knowledge (e.g. physics and chemistry) still important in AI for protein science? Why or why not?

References

- Senior et al. "Improved protein structure prediction using potentials from deep learning", Nature, Vol. 577, pp. 706-710, 2020 (**AlphaFold 1**).
- Jumper et al. "Highly accurate protein structure prediction with AlphaFold", Nature, Vol. 596, pp. 583-589, 2021 (**AlphaFold 2**).
- Abramson et al. "Accurate structure prediction of biomolecular interactions with AlphaFold 3", Nature, Vol. 630, pp. 493-500, 2024 (**AlphaFold 3**).
- https://cvpr2022-tutorial-diffusion-models.github.io/ (CVPR 2022 Tutorial on Denoising diffusion-based generative modeling)
- Song et al. "Score-based generative modeling through stochastic differential equations", ICLR 2021.
- Karras et al. "Elucidating the design space of diffusion-based generative models", NeurIPS 2022.